AIN SHAMS · INTERNAL MEDICINE COURSE 2025–2026

Exam Case Bank

42 exam-style clinical cases with 238 questions and full marking schemes. Each case has 5–10 questions, a criteria table where one is needed, and a drug-class table at the end. Built from your lecture slides, handouts and the revision guide, weighted towards what is most likely to be examined.

A topics → doses expected in the answer

B topics → main lines + referral

C topics → diagnosis & DD

Cardiology	Tropical	Chest	Haematology	Immunology
12 cases	11 cases	8 cases	8 cases	3 cases

0 How to use the case bank

Cover the green marking scheme, answer out loud or on paper, then mark yourself

Each case looks like this

- **Header:** case number, a title that doesn't give the diagnosis away, the study-guide level and the total marks.
- **Stem:** the scenario, with vital signs and results in grey chips.
- **Questions** (Q1, Q2…) with the marks for each one on the right.
- **Marking scheme** (green bar) right under each question. Each bullet is one mark point; the mark is on the right of the bullet.
- **Key criteria table** (amber) whenever an answer depends on a scoring system or criteria (Jones, Wells, CURB-65, Light's, Duke…).
- **Drugs in this case** (slate table) at the end: every drug named in the case, its **class**, and the one point examiners ask about.

★ HOW EXAMINERS GIVE MARKS

- **Drug + dose + route + frequency/duration** = full mark. A drug name alone gets half.
- "Investigations": the **first** / bedside test first, then the confirmatory test, then the rest.
- "Diagnosis" means a full diagnosis: *what* + *where* + *cause* + *complications* (e.g. "acute inferior STEMI complicated by complete heart block").
- Don't list everything you know. One wrong dangerous drug can cancel a correct answer.

Case index

Subject	Cases
Cardiology	1 STEMI · 2 CCS → UA · 3 resistant HTN · 4 HTN emergency & dissection · 5 chronic → acute HF · 6 SVT · 7 AF · 8 cardiac arrest · 9 RF → MS · 10 IE · 11 PE · 12 pericarditis → tamponade
Chest	13 COPD exacerbation · 14 acute severe asthma · 15 CAP · 16 pulmonary TB · 17 pleural effusion · 18 pneumothorax · 19 IPF · 20 lung cancer
Tropical	21 amoebic liver abscess · 22 dysentery & cholera · 23 brucellosis · 24 meningitis · 25 viral hepatitis · 26 infectious mononucleosis · 27 FUO · 28 schistosomiasis · 29 falciparum malaria · 30 septic shock · 31 HIV & needle-stick
Haematology	32 iron deficiency vs thal trait · 33 B12 deficiency · 34 ITP · 35 haemophilia vs vWD · 36 haemolytic anaemia · 37 β-thalassaemia major · 38 transfusion reaction & DIC · 39 thrombophilia
Immunology	40 anaphylaxis · 41 angioedema · 42 chronic urticaria & drug allergy

⚠ DOSES

Doses follow your lecture slides where the slides give them; otherwise they follow current standard guidelines (ESC, GINA, GOLD, WHO). Where the two differ, the scheme gives the lecture answer and notes the guideline.

Psychiatry, dermatology and STD/andrology have no cases, as you asked. Use the revision guide for those.

1 Cardiology cases

12 cases · 75 questions | STEMI · CCS/UA · HTN · dissection · HF · SVT · AF · arrest · RF/MS · IE · PE · pericardium

Case 1 Crushing chest pain at dawn

A · 25 marks

A 58-year-old male smoker with hypertension and type 2 diabetes arrives at the ER **4 hours** after the onset of severe, persistent retrosternal chest pain radiating to the left shoulder and arm, lasting more than 30 minutes, with sweating, pallor and nausea. The nearest PCI-capable hospital is 45 minutes away.

BP 110/70 | HR 52, regular | RR 20 | SpO2 95% air | Chest: clear | ECG: ST elevation 3 mm in II, III, aVF; ST depression I, aVL

Q1 What is the immediate bedside investigation, and what is your diagnosis? **3 marks**

Q2 Give the immediate medications with doses. **5 marks**

Q3 What is the definitive treatment and how do you choose between the options here? **4 marks**

Q4 Mention 4 absolute contraindications to thrombolysis. **2 marks**

Q5 One hour later the HR falls to 38/min, BP 85/50. ECG: P waves march through regular broad QRS complexes with no relation. Diagnosis and management? **4 marks**

Q6 Which cardiac markers would you follow, and why is CK-MB still useful? **3 marks**

Q7 List his discharge (secondary prevention) medications. **4 marks**

Case 1 Crushing chest pain at dawn

A · 25 marks

Q1 What is the immediate bedside investigation, and what is your diagnosis? **3 marks**

MARKING SCHEME

- **12-lead ECG within 10 minutes** of arrival — not troponin (don't wait for markers to treat a STEMI) 1
- **Acute inferior STEMI** (ST↑ ≥2 contiguous inferior leads with reciprocal ST↓ in I, aVL) 1
- Likely **right coronary artery** occlusion; bradycardia suggests AV-node ischaemia 1

Q2 Give the immediate medications with doses. **5 marks**

MARKING SCHEME

- **Aspirin 300 mg chewed** (loading), then 75–100 mg daily 1
- **Clopidogrel 600 mg** loading (or ticagrelor 180 mg), then 75 mg daily (ticagrelor 90 mg bd) 1
- **Anticoagulant:** UFH 70–100 IU/kg IV bolus (or enoxaparin 0.5 mg/kg IV then 1 mg/kg SC bd) 1
- **Analgesia:** morphine 2–5 mg IV slowly + antiemetic (metoclopramide 10 mg IV) 1
- O2 only if SpO2 <90% (here not needed) · **avoid nitrates** if RV infarction/hypotension suspected 1

Q3 What is the definitive treatment and how do you choose between the options here? **4 marks**

MARKING SCHEME

- Urgent **reperfusion** — symptoms <12 h 1
- **Primary PCI is preferred** if it can be done **≤120 min** from diagnosis → transfer now (45 min away = achievable) 1
- If PCI not achievable in 120 min → **fibrinolysis within 10 min** of diagnosis (e.g. streptokinase 1.5 MU IV over 60 min, or tenecteplase/alteplase) 1
- After lysis → transfer for angiography 2–24 h (rescue PCI immediately if lysis fails: <50% ST resolution at 60–90 min) 1

Q4 Mention 4 absolute contraindications to thrombolysis. **2 marks**

MARKING SCHEME (½ EACH, ANY 4)

- Previous intracranial haemorrhage (any time)
- Ischaemic stroke in last 6 months
- CNS neoplasm / AVM
- Major trauma, surgery or head injury <1 month
- GI bleeding <1 month
- Known bleeding disorder
- **Aortic dissection**
- Non-compressible punctures <24 h

Q5 One hour later the HR falls to 38/min, BP 85/50. ECG: P waves march through regular broad QRS complexes with no relation. Diagnosis and management? **4 marks**

MARKING SCHEME

- **Complete (3rd-degree) heart block** complicating inferior STEMI 1
- **Atropine 0.5–1 mg IV**, repeat every 3–5 min up to 3 mg (block at AV-node level in inferior MI) 1
- If no response → **transcutaneous pacing** → temporary transvenous pacemaker; adrenaline/isoprenaline infusion as bridge 1
- Reperfusion (PCI) usually resolves it; inferior-MI CHB is often transient → PPM rarely needed 1

Q6 Which cardiac markers would you follow, and why is CK-MB still useful? **3 marks**

MARKING SCHEME

- **Troponin I/T**: marker of choice — rises 2–4 h, peaks 10–24 h, stays 5–14 days 1
- **CK-MB**: normalises in 2–3 days → detects **re-infarction**; level ∝ infarct size 1
- Myoglobin rises earliest (1–2 h) — good **negative** predictor, not specific 1

Q7 List his discharge (secondary prevention) medications. **4 marks**

MARKING SCHEME

- **DAPT**: aspirin 75–100 mg + clopidogrel 75 mg (or ticagrelor 90 mg bd) for **12 months** 1
- **High-intensity statin**: atorvastatin 80 mg (LDL target <55 mg/dL) 1
- **β-blocker** (bisoprolol) + **ACE inhibitor** (ramipril; diabetes, HTN, EF ≤40%) ± MRA if EF ≤40% 1
- Stop smoking, cardiac rehab, control DM & BP, SL nitrate PRN 1

KEY CRITERIA · STEMI TERRITORIES & REPERFUSION TIMES

Territory	Leads	Artery	Timeline	Target
Inferior	II, III, aVF (+ V4R for RV)	RCA (~80%)	ECG	≤10 min of first contact
Anterior / septal	V1–V4	LAD	Primary PCI	≤120 min from diagnosis (symptoms ≤12 h)
Lateral	I, aVL, V5–V6	LCx	Fibrinolysis	≤10 min of diagnosis if PCI too late
Posterior	ST↓ + tall R V1–V3 (V7–V9 ↑)	LCx/RCA	Angio after lysis	2–24 h

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Aspirin	Antiplatelet — irreversible COX-1 inhibitor	300 mg chewed loading
Clopidogrel / ticagrelor	Antiplatelet — **P2Y12 (ADP-receptor) inhibitor**	Part of DAPT for 12 months
Unfractionated heparin	Anticoagulant — indirect thrombin/Xa inhibitor (via antithrombin)	Monitor aPTT
Enoxaparin	Anticoagulant — LMWH (mainly anti-Xa)	No routine monitoring
Streptokinase / alteplase / tenecteplase	Fibrinolytics (thrombolytics) — plasminogen activators	Streptokinase: antigenic, give once
Morphine	Opioid analgesic	Pain + anxiolysis; give antiemetic
Atropine	Antimuscarinic (parasympatholytic)	Works at AV-node level only
Atorvastatin	Lipid-lowering — HMG-CoA reductase inhibitor (statin)	High intensity after ACS
Bisoprolol	β1-selective blocker	Avoid acutely in bradycardia/HF/shock
Ramipril	ACE inhibitor	Remodelling; check K^+, creatinine

Case 2 Exertional pain that changed its pattern

A · 20 marks

A 57-year-old woman with diabetes and hypertension has had retrosternal tightness on climbing stairs for 1 year, relieved by rest within minutes. Her rest ECG and echo were normal at that time. Over the last week the pain has become more severe and longer, and today it occurred **at rest** and did not settle with sublingual nitrate.

BP 150/90 | HR 92 | ECG: 2 mm horizontal ST depression V4–V6 | Troponin at 0 h and 3 h: normal

Q1 Classify her original chest pain and justify. **2 marks**

Q2 When she first presented (normal rest ECG & echo), what was the most important next test, and what would you have given? **4 marks**

Q3 What is the diagnosis now? Justify. **3 marks**

Q4 Outline her immediate management with doses. **5 marks**

Q5 When should she have coronary angiography? **3 marks**

Q6 Give 3 non-cardiac causes of chest pain you would consider and one clue for each. **3 marks**

Case 2 Exertional pain that changed its pattern

A · 20 marks

Q1 Classify her original chest pain and justify. **2 marks**

MARKING SCHEME

- **Typical angina** (chronic coronary syndrome) 1
- Meets all 3: substernal characteristic discomfort · provoked by exertion · relieved by rest/nitrate 1

Q2 When she first presented (normal rest ECG & echo), what was the most important next test, and what would you have given? **4 marks**

MARKING SCHEME

- **Stress test** — exercise ECG if able to exercise (else dobutamine echo / vasodilator SPECT / stress MRI); coronary CTA alternative 1
- Anti-anginal: **β-blocker** (bisoprolol 5–10 mg) ± CCB (amlodipine 5–10 mg), **SL GTN 0.4–0.5 mg PRN** 1
- Event prevention: **aspirin 75–100 mg** + **high-intensity statin** 1
- Coronary angiography if symptoms persist despite OMT or high-risk stress test 1

Q3 What is the diagnosis now? Justify. **3 marks**

MARKING SCHEME

- **Unstable angina** (NSTE-ACS) 1
- Crescendo + rest pain + nitrate-resistant = unstable; ST depression = subendocardial ischaemia 1
- **Serial troponins normal** → not NSTEMI; normal ECG would not exclude ACS either 1

Q4 Outline her immediate management with doses. **5 marks**

MARKING SCHEME

- Admit to CCU, monitor, IV line, analgesia 1
- **Aspirin 300 mg** load → 75–100 mg **+ P2Y12 inhibitor** (ticagrelor 180 mg → 90 mg bd, or clopidogrel 300–600 mg → 75 mg) 1
- **Anticoagulation**: fondaparinux 2.5 mg SC daily or enoxaparin 1 mg/kg SC bd 1
- Anti-ischaemic: IV/SL nitrate, β-blocker (if no HF/shock), statin 1
- Control BP & glucose; **no fibrinolysis** in NSTE-ACS 1

Q5 When should she have coronary angiography? **3 marks**

MARKING SCHEME

- **Immediate (<2 h)** if very high risk: refractory pain, haemodynamic instability/shock, life-threatening arrhythmia, acute HF 1
- **Early (<24 h)** if high risk: rising troponin (NSTEMI), dynamic ST changes, GRACE >140 1
- Her case (dynamic ST↓, rest pain) → invasive strategy ≤24 h, then PCI or CABG (multivessel + DM → CABG favoured) 1

Q6 Give 3 non-cardiac causes of chest pain you would consider and one clue for each. **3 marks**

MARKING SCHEME (1 EACH, ANY 3)

- Oesophageal spasm/GERD — burning, related to food, **relieved by NTG too**
- Musculoskeletal (costochondritis) — **reproducible on palpation**
- PE — pleuritic, dyspnoea, DVT risk · Pneumothorax — sudden, absent breath sounds
- Herpes zoster — burning dermatomal pain before vesicles · biliary/pancreatic — epigastric, tender abdomen

KEY CRITERIA · ANGINA CLASSIFICATION & ACS SPECTRUM

Typical angina	Atypical	Non-cardiac	UA	NSTEMI	STEMI
3/3 features	2/3	≤1/3	Troponin normal; ST↓/T inversion/normal	Troponin ↑; ST↓/T inversion/normal	ST↑ ≥2 contiguous leads or new LBBB

DRUGS IN THIS CASE

Drug	Class	Key point
Glyceryl trinitrate	Nitrate — venodilator (NO donor)	Headache; contraindicated with PDE5 inhibitors
Ticagrelor	P2Y12 inhibitor (reversible)	Dyspnoea side effect; bd dosing
Fondaparinux	Anticoagulant — synthetic indirect factor Xa inhibitor	Preferred in NSTE-ACS (less bleeding)
Amlodipine	Dihydropyridine calcium-channel blocker	Ankle oedema
Isosorbide mononitrate	Long-acting nitrate	Nitrate-free interval (tolerance)

Case 3 Three drugs and still 165/110

A · 20 marks

A 60-year-old man (BMI 40) complains of daytime sleepiness, snoring, easy fatigue and exertional dyspnoea. Hypertensive for 5 years on **amlodipine 10 mg + valsartan 320 mg + hydrochlorothiazide 25 mg**, which he says he takes regularly. Office BP is always about 165/110.

K^+ 3.1 mmol/L | Na^+ 144 | Creatinine 1.0 | ECG: LVH with strain

Q1 Define resistant hypertension. Does he fulfil it yet? **3 marks**

Q2 Mention 4 causes of pseudo-resistance. **2 marks**

Q3 Give your differential diagnosis for true resistance in this man. **4 marks**

Q4 What investigations would you request? **4 marks**

Q5 What is his BP target and what would you add? **4 marks**

Q6 List 6 precautions when measuring his blood pressure. **3 marks**

Case 3 Three drugs and still 165/110

A · 20 marks

Q1 Define resistant hypertension. Does he fulfil it yet? **3 marks**

MARKING SCHEME

- BP ≥140/90 despite lifestyle measures + **maximally tolerated doses of a thiazide, a RAS blocker and a CCB** — 1
- Must be **confirmed by out-of-office BP (24-h ABPM)** — 1
- Pseudo-resistance must be excluded first → not yet confirmed — 1

Q2 Mention 4 causes of pseudo-resistance. **2 marks**

MARKING SCHEME (½ EACH)

- Inaccurate measurement (small cuff in obese arm!)
- White-coat effect
- Poor adherence to drugs / lifestyle
- Suboptimal drug choice or doses

Q3 Give your differential diagnosis for true resistance in this man. **4 marks**

MARKING SCHEME

- **Obstructive sleep apnoea** (obesity, snoring, sleepiness) — 1
- **Primary aldosteronism** — **hypokalaemia** on thiazide; most common secondary cause — 1
- Renal parenchymal disease / renal artery stenosis; Cushing; phaeochromocytoma; thyroid — 1
- Volume overload (salt intake), offending drugs (NSAIDs, steroids, decongestants, liquorice) — 1

Q4 What investigations would you request? **4 marks**

MARKING SCHEME

- **24-h ABPM** to confirm; correct large cuff — 1
- **Plasma aldosterone : renin ratio** (correct K^+ first) — 1
- **Polysomnography** (sleep study) for OSA — 1
- Urine ACR, renal US/renal artery Doppler, echo (LVH), fundoscopy; metanephrines/cortisol if clues — 1

Q5 What is his BP target and what would you add? **4 marks**

MARKING SCHEME

- Target **120–129 / 70–79 mmHg** (ESC 2024) if tolerated — 1
- Add **spironolactone 25–50 mg daily** (4th-line; also corrects K^+, treats aldosteronism) — 1
- Alternatives: β-blocker, α-blocker (doxazosin), amiloride, eplerenone; treat OSA with **CPAP** — 1
- Weight loss, salt <5 g/day, DASH diet, exercise, stop alcohol; refer to HTN specialist — 1

Q6 List 6 precautions when measuring his blood pressure. **3 marks**

MARKING SCHEME (½ EACH, ANY 6)

- Seated 3–5 min, back supported
- Cuff at heart level, not over clothes
- Bladder length ≥80%, width ≥40% of arm circumference
- Both arms first visit (difference >20 → investigate)
- ≥2 readings 1 min apart
- No caffeine/smoking 1 h before; empty bladder
- Palpatory method first (auscultatory gap)
- Deflate 2–3 mmHg/beat; no talking

KEY CRITERIA · ESC 2024 BP CATEGORIES & WHO GETS DRUGS

Category	Office BP	Action
Non-elevated	<120 / <70	Lifestyle for prevention
Elevated	120–139 / 70–89	Lifestyle 3 months; drug if SBP 130–139 + high risk (CVD, DM, CKD, HMOD)
Hypertension	≥140 / ≥90	Lifestyle + immediate drug (single-pill combination)
Out-of-office HTN	ABPM/home ≥135/85	Confirms diagnosis, excludes white coat

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Amlodipine	Dihydropyridine CCB	"C" of A+C+D
Valsartan	Angiotensin II receptor blocker (ARB)	"A"; no cough (unlike ACEi)
Hydrochlorothiazide	Thiazide diuretic	"D"; causes ↓K^+, ↑glucose, ↑urate
Spironolactone	Aldosterone (mineralocorticoid-receptor) antagonist — K^+-sparing diuretic	4th drug; gynaecomastia, ↑K^+
Doxazosin	α1-adrenergic blocker	Postural hypotension; useful in BPH

Case 4 Tearing pain through to the back

A / B · 20 marks

A 62-year-old poorly controlled hypertensive man suddenly develops severe tearing chest pain radiating to the back between the shoulder blades. He is sweaty and anxious.

BP right arm 220/140 | BP left arm 170/100 | HR 110 | Early diastolic murmur, left sternal edge | ECG: sinus tachycardia, LVH | CXR: widened mediastinum

Q1 What is the most likely diagnosis? Give 4 supporting points. **3 marks**

Q2 What is your FIRST step in management, with targets? **4 marks**

Q3 How would you confirm the diagnosis (a) if he is stable and (b) if his BP becomes inaudible? **3 marks**

Q4 Classify aortic dissection and give the management of each type. **4 marks**

Q5 His ECG later shows inferior ST elevation. The junior suggests thrombolysis. Comment. **2 marks**

Q6 Distinguish hypertensive emergency from urgency and state the general BP-lowering rule. **4 marks**

Case 4 Tearing pain through to the back

A / B · 20 marks

Q1 What is the most likely diagnosis? Give 4 supporting points. **3 marks**

MARKING SCHEME

- **Acute aortic dissection** (likely type A — involves ascending aorta) 1
- Tearing pain to back; uncontrolled HTN; **inter-arm BP difference >20** 1
- **New AR murmur** (aortic-root dilatation); widened mediastinum 1

Q2 What is your FIRST step in management, with targets? **4 marks**

MARKING SCHEME

- **IV β-blocker first**: esmolol 0.5 mg/kg bolus → 0.05–0.3 mg/kg/min, or IV labetalol 20 mg → infusion 1
- Targets: **SBP <120 mmHg** and **HR <60/min** immediately (↓ aortic shear stress) 1
- Add vasodilator (nitroprusside) only **after** β-blockade (avoid reflex tachycardia); diltiazem if BB contraindicated 1
- **IV morphine** for pain (also ↓ sympathetic drive); ICU, 2 large IV lines, crossmatch 1

Q3 How would you confirm the diagnosis (a) if he is stable and (b) if his BP becomes inaudible? **3 marks**

MARKING SCHEME

- (a) **CT angiography of the aorta** (or MRI) 1
- (b) Unstable patient is **never sent to radiology** → **bedside TTE/TEE** (flap, AR, effusion) 1
- ECG to look for coronary involvement; notify cardiothoracic theatre 1

Q4 Classify aortic dissection and give the management of each type. **4 marks**

MARKING SCHEME

- **Stanford A**: involves ascending aorta (DeBakey I, II) → **emergency surgery** 1
- **Stanford B**: descending only (DeBakey III) → **medical** (pain + BP/HR control, CT/MRI surveillance) 1
- Complicated B (rupture, malperfusion, expansion) → endovascular stent (TEVAR) 1
- Risk factors: HTN, Marfan/Ehlers-Danlos, bicuspid valve, coarctation, pregnancy, cocaine 1

Q5 His ECG later shows inferior ST elevation. The junior suggests thrombolysis. Comment. **2 marks**

MARKING SCHEME

- Dissection extending into the **right coronary ostium** → inferior STEMI 1
- Thrombolysis is **absolutely contraindicated** (fatal rupture/tamponade) → surgery 1

Q6 Distinguish hypertensive emergency from urgency and state the general BP-lowering rule. **4 marks**

MARKING SCHEME

- **Emergency**: BP >180/120 **+ acute organ damage** → ICU + IV drugs 1
- **Urgency**: >180/120 without organ damage → oral drugs, lower over 24–48 h 1
- Lower **MAP by 20–25% in first 1–2 h** (except dissection SBP <120; ACS/pulmonary oedema SBP <140) 1
- **Sublingual nifedipine is contraindicated**; IV labetalol/nitroprusside best; MgSO4 in eclampsia 1

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Esmolol	Ultra-short-acting β1-blocker (IV)	Titratable; first in dissection
Labetalol	Combined α + β blocker	Also safe in pregnancy
Sodium nitroprusside	Direct arterial + venous vasodilator (NO donor)	Thiocyanate/cyanide toxicity >24 h
Diltiazem	Non-dihydropyridine CCB	Rate control if BB contraindicated
Morphine	Opioid analgesic	Reduces sympathetic drive

Case 5 Breathless, swollen and waiting for medicines

A · 26 marks

Mr O, 54, smoker (20 pack-years), diabetic and hypertensive (on bisoprolol), has had exertional angina for 3 years and refused coronary angiography. For 2 weeks he has had progressive dyspnoea, orthopnoea, cough with frothy sputum and leg swelling.

BP 100/70 | HR 110 regular | RR 20 | ↑JVP | Bilateral LL oedema, moderate ascites | Bibasal fine crepitations | Apex 6th ICS AAL, S3 gallop | Pansystolic apical murmur | Creatinine 2.0 | Hb 9 g/dL | K^+ 5.6 | Echo: EF 35%, moderate MR

Q1 Give the full diagnosis. **3 marks**

Q2 Classify his signs into left-sided, right-sided and low-output features. **3 marks**

Q3 Which investigations would you order and why? **4 marks**

Q4 Outline long-term drug therapy with doses. What must you watch given his K^+ and creatinine? **6 marks**

Q5 Which drugs improve survival and which only improve symptoms? **2 marks**

Q6 While waiting for insurance to buy his medicines he suddenly becomes severely breathless. BP 85/55, HR 150, cold peripheries, lungs congested. What are the possible precipitants of decompensation (list 5)? **3 marks**

Q7 What is his current state and its management? **5 marks**

Case 5 Breathless, swollen and waiting for medicines

A · 26 marks

Q1 Give the full diagnosis. **3 marks**

MARKING SCHEME

- **Congestive (biventricular) heart failure with reduced EF (HFrEF, EF 35%)**, NYHA III–IV — 1
- Cause: **ischaemic cardiomyopathy** (old IHD) with functional MR — 1
- Comorbidities: CKD/cardiorenal, anaemia, hyperkalaemia, DM, HTN, smoker — 1

Q2 Classify his signs into left-sided, right-sided and low-output features. **3 marks**

MARKING SCHEME

- **Left (pulmonary venous congestion)**: dyspnoea, orthopnoea, frothy sputum, crepitations, S3 — 1
- **Right (systemic congestion)**: ↑JVP, LL oedema, ascites, (tender hepatomegaly) — 1
- **Low output**: fatigue, tachycardia, low-normal BP, renal impairment — 1

Q3 Which investigations would you order and why? **4 marks**

MARKING SCHEME

- **Echocardiography** — the key test: EF, valves, wall motion — 1
- **BNP / NT-proBNP** — supports diagnosis, prognosis (low value rules HF out) — 1
- ECG (ischaemia, arrhythmia, LVH) · CXR (cardiomegaly, upper-lobe diversion, Kerley B, effusions) — 1
- CBC, renal, electrolytes, TSH, HbA1c, lipids, iron studies · **coronary angiography** for the cause — 1

Q4 Outline long-term drug therapy with doses. What must you watch given his K^+ and creatinine? **6 marks**

MARKING SCHEME

- Symptom relief: **furosemide 40–80 mg/day** to dry weight — 1
- **ACE inhibitor** (ramipril 1.25 mg → 10 mg) or ARB or **ARNI** sacubitril/valsartan — 1
- **β-blocker**: continue/convert to bisoprolol 1.25 → 10 mg once euvolaemic — 1
- **MRA**: spironolactone 25 mg — **hold now (K^+ 5.6)**; start once K^+ <5.0 — 1
- **SGLT2 inhibitor** dapagliflozin 10 mg (also helps DM/CKD) — 1
- Monitor K^+ and creatinine 1–2 weeks after each change; correct anaemia (IV iron); revascularise — 1

Q5 Which drugs improve survival and which only improve symptoms? **2 marks**

MARKING SCHEME

- **Mortality benefit**: ACEi/ARB/ARNI, β-blocker, MRA, SGLT2 inhibitor — 1
- **Symptoms only**: loop diuretics, digoxin, (thiazides) — 1

Q6 While waiting for insurance to buy his medicines he suddenly becomes severely breathless. BP 85/55, HR 150, cold peripheries, lungs congested. What are the possible precipitants of decompensation (list 5)? **3 marks**

MARKING SCHEME (ANY 5; ½ EACH, MAX 3)

- **Non-adherence** to drugs/salt/fluid (here!)
- ACS / ischaemia
- Arrhythmia (AF, VT)
- Infection (pneumonia, IE)
- Uncontrolled HTN
- Drugs: NSAIDs, steroids, verapamil
- Alcohol, liquorice
- Anaemia, thyroid, PE

Q7 What is his current state and its management? **5 marks**

MARKING SCHEME

- **Cardiogenic shock** (cold & wet acute HF) — check ECG for arrhythmia (HR 150 → if VT/rapid AF → **DC shock**) — 1
- O2 / NIV → intubation if failing; ICU; arterial line, urine catheter, lactate — 1
- **Inotrope dobutamine** 2–20 µg/kg/min ± **vasopressor norepinephrine** 0.05–0.5 µg/kg/min for MAP ≥65 — 1
- IV furosemide once perfusion improves; **avoid nitrates** (hypotension); stop β-blocker temporarily — 1
- Treat cause: urgent angiography/PCI if ACS; mechanical support (IABP/ECMO) if refractory — 1

KEY CRITERIA · NYHA & EF CLASSIFICATION

NYHA I	II	III	IV	HFrEF	HFmrEF	HFpEF
No limitation	Slight — symptoms on ordinary activity	Marked — less than ordinary	At rest	EF ≤40%	41–49%	≥50%

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Furosemide	Loop diuretic	Symptoms only; ↓K^+
Ramipril	ACE inhibitor	Mortality ↓; cough, ↑K^+, angioedema
Sacubitril/valsartan	ARNI — neprilysin inhibitor + ARB	Replaces ACEi (36 h washout)
Bisoprolol / carvedilol	β-blocker (bisoprolol β1-selective; carvedilol α+β)	Start low, go slow
Spironolactone	Mineralocorticoid-receptor antagonist (MRA)	Hold if K^+ >5.0–5.5
Dapagliflozin	SGLT2 inhibitor	4th pillar; genital infections
Digoxin	Cardiac glycoside (Na/K-ATPase inhibitor)	Symptoms, AF rate; toxicity ↑ with ↓K^+
Dobutamine	Inotrope — β1-agonist	Cardiogenic shock
Norepinephrine	Vasopressor — α1 (> β1) agonist	Maintain MAP

Case 6 Sudden racing heart in a young woman

A · 15 marks

A 24-year-old woman presents with sudden-onset regular palpitations for 40 minutes, mild dizziness, no chest pain. She has had similar shorter episodes that stopped by themselves.

BP 115/75 | HR 186 regular | Alert | ECG: regular narrow-QRS tachycardia, no visible P waves

Q1 What is the rhythm and how do you decide on treatment? **3 marks**

Q2 Describe the first-line steps with doses. **5 marks**

Q3 Adenosine fails. What next? **3 marks**

Q4 After adenosine, the rate slows and saw-tooth waves appear at 300/min with 2:1 then 4:1 conduction. Interpret. **2 marks**

Q5 After conversion, the ECG shows short PR and a delta wave. What is it and which drugs should be avoided if she later develops AF? **2 marks**

Case 6 Sudden racing heart in a young woman

A · 15 marks

Q1 What is the rhythm and how do you decide on treatment? **3 marks**

MARKING SCHEME

- **Paroxysmal supraventricular tachycardia** (AVNRT / AVRT) — regular, narrow, absent/retrograde P — 1
- First decide **haemodynamic stability** (shock, syncope, ischaemic pain, acute HF) — 1
- She is **stable** → vagal → drugs; unstable → synchronised DC — 1

Q2 Describe the first-line steps with doses. **5 marks**

MARKING SCHEME

- **Vagal manoeuvres**: modified Valsalva, carotid sinus massage (one side, no bruit) — 1
- **Adenosine 6 mg rapid IV bolus** + saline flush, large proximal vein (antecubital/jugular) — 1
- No response in 2 min → **12 mg** → further **12 mg** — 1
- ECG monitoring; warn of flushing, chest tightness, sense of doom (t½ seconds) — 1
- Avoid adenosine in asthma (bronchospasm) → use verapamil — 1

Q3 Adenosine fails. What next? **3 marks**

MARKING SCHEME

- **Verapamil 5 mg IV over 1–2 min**, repeat 5–10 mg after 10–15 min — 1
- or **propranolol 1 mg IV over 1–2 min**, repeat every 5 min to max 5 mg — 1
- All drugs fail → **synchronised DC** 100 → 200 J biphasic (150 → 360 J monophasic) — 1

Q4 After adenosine, the rate slows and saw-tooth waves appear at 300/min with 2:1 then 4:1 conduction. Interpret. **2 marks**

MARKING SCHEME

- This is **atrial flutter** (2:1 at ~150) unmasked — AV-nodal block doesn't terminate it — 1
- Manage as AF: rate control / cardioversion + anticoagulation assessment; ablation curative — 1

Q5 After conversion, the ECG shows short PR and a delta wave. What is it and which drugs should be avoided if she later develops AF? **2 marks**

MARKING SCHEME

- **WPW syndrome** (accessory pathway, pre-excitation) → refer for **catheter ablation** — 1
- Pre-excited AF: avoid AV-nodal blockers (**adenosine, verapamil, digoxin, β-blockers**) → DC or procainamide/flecainide — 1

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Adenosine	Purine nucleoside — transient AV-node blocker (A1 agonist)	t½ seconds; avoid in asthma
Verapamil	Non-DHP CCB (class IV antiarrhythmic)	Never with IV β-blocker or in WCT
Propranolol	Non-selective β-blocker (class II)	Avoid in asthma
Procainamide / flecainide	Class Ia / Ic antiarrhythmics (Na-channel blockers)	Used in pre-excited AF

Case 7 Irregular palpitations — how long has it been?

A · 22 marks

A 70-year-old woman, known rheumatic mitral stenosis and hypertension, presents with palpitations and dyspnoea. She does not know when they started. She had a TIA 1 year ago.

BP 130/80 | HR ~150, irregularly irregular | No signs of shock | ECG: no P waves, fibrillatory baseline, narrow QRS | Weight 60 kg

Q1 Diagnosis and immediate investigations? **3 marks**

Q2 Why is urgent cardioversion NOT the right choice for her? **2 marks**

Q3 How do you control her rate? Give target and drugs. **4 marks**

Q4 What anticoagulant does she need, how and for how long before elective cardioversion? **4 marks**

Q5 Calculate her CHA_2DS_2-VASc score. **2 marks**

Q6 A different patient (no structural heart disease) presents with AF of 10 hours' duration. How would you cardiovert pharmacologically, and how would you change this if he had structural heart disease? **4 marks**

Q7 If she suddenly becomes hypotensive (BP 70/40) with HR 170, what do you do, and what must you exclude first? **3 marks**

Case 7 Irregular palpitations — how long has it been?

A · 22 marks

Q1 Diagnosis and immediate investigations? **3 marks**

MARKING SCHEME

- **Atrial fibrillation with rapid ventricular response**, unknown duration, on rheumatic MS — 1
- ECG, **TSH**, electrolytes (K^+, Mg^{2+}), CBC, renal, glucose; look for a systemic cause (sepsis, PE, bleeding) — 1
- **Echo** (MS severity, LA size, LV function); TEE before any cardioversion — 1

Q2 Why is urgent cardioversion NOT the right choice for her? **2 marks**

MARKING SCHEME

- Duration **>48 h / unknown**, previous thromboembolism and **rheumatic MS** → high LA-thrombus risk — 1
- Cardioversion could dislodge clot → stroke; she is stable → rate control + anticoagulation first — 1

Q3 How do you control her rate? Give target and drugs. **4 marks**

MARKING SCHEME

- Target resting HR **80–110/min** (lenient) — 1
- Preserved LV: **β-blocker** (IV metoprolol 2.5–5 mg / oral bisoprolol) or **verapamil/diltiazem** — 1
- Reduced LV / HF: **digoxin** 0.25–0.5 mg IV load or **amiodarone** IV — 1
- Treat precipitants (fever, thyroid, anaemia); in MS slower rate prolongs diastolic filling — 1

Q4 What anticoagulant does she need, how and for how long before elective cardioversion? **4 marks**

MARKING SCHEME

- Moderate–severe rheumatic MS → **warfarin** (DOACs not indicated in moderate–severe MS / mechanical valves) — 1
- Start **heparin bridge** (LMWH 1 mg/kg bd or UFH) + warfarin, **INR 2–3** — 1
- At least **3–4 weeks** of therapeutic OAC before cardioversion (or TEE to exclude LA thrombus) — 1
- Continue ≥4 weeks after cardioversion — **lifelong** in her (MS + prior TIA) — 1

Q5 Calculate her CHA_2DS_2-VASc score. **2 marks**

MARKING SCHEME

- HTN 1 + age 65–74 1 + female 1 + TIA 2 = **5** — 1
- ≥2 (men) / ≥3 (women) → OAC indicated (and MS alone mandates it) — 1

Q6 A different patient (no structural heart disease) presents with AF of 10 hours' duration. How would you cardiovert pharmacologically, and how would you change this if he had structural heart disease? **4 marks**

MARKING SCHEME

- <48 h → **urgent cardioversion** (drug or DC) with heparin cover — 1
- No structural HD: **propafenone 600 mg orally** (450–600 mg) or flecainide 200–300 mg (pill-in-the-pocket) — 1
- Structural HD/IHD: **amiodarone 5–7 mg/kg IV over 1–2 h → 50 mg/h** (max 1.2 g/24 h) — 1
- Afterwards: echo, TSH, CBC, electrolytes; long-term OAC by CHA_2DS_2-VASc — 1

Q7 If she suddenly becomes hypotensive (BP 70/40) with HR 170, what do you do, and what must you exclude first? **3 marks**

MARKING SCHEME

- **UFH 80 IU/kg bolus** then **synchronised DC shock** — 1
- Only if AF itself is the cause of shock (usually HR >150) — 1
- Exclude other shock causes: **GI bleeding/melaena, massive PE, sepsis, cardiogenic shock** — 1

KEY CRITERIA · CHA_2DS_2-VASC

C	H	A_2	D	S_2	V	A	Sc
CHF / LV dysfunction 1	Hypertension 1	Age ≥75 = 2	Diabetes 1	Stroke/TIA/embolism 2	Vascular disease 1	Age 65–74 = 1	Female sex 1

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Metoprolol / bisoprolol	β1-selective blocker (class II)	First-line rate control
Diltiazem / verapamil	Non-DHP CCB (class IV)	Avoid in HFrEF
Digoxin	Cardiac glycoside	Rate control at rest; HF
Amiodarone	Class III antiarrhythmic (K-channel blocker)	Thyroid, lung, liver toxicity; phlebitis
Propafenone / flecainide	Class Ic antiarrhythmics	Not in IHD/structural HD
Warfarin	Oral anticoagulant — vitamin K antagonist	INR 2–3; needed in MS
Apixaban / rivaroxaban / dabigatran	DOACs — direct Xa inhibitors / direct thrombin inhibitor	Not in mod–severe MS / mechanical valve
Enoxaparin / UFH	Anticoagulants — LMWH / unfractionated heparin	Bridging

Case 8 Collapse on the ward

A · 18 marks

A 65-year-old man admitted with NSTEMI suddenly collapses on the ward. He is unresponsive, not breathing normally, and no carotid pulse is felt. The monitor shows a chaotic, irregular undulating baseline with no identifiable complexes.

Q1 Define cardiac arrest and name this rhythm. **2 marks**

Q2 Describe your immediate actions. **5 marks**

Q3 Which drugs, at what doses and when? **3 marks**

Q4 List the reversible causes you must look for. **3 marks**

Q5 How would management differ if the monitor showed a normal-looking sinus rhythm but there was still no pulse? **3 marks**

Q6 Compare BLS and ACLS. **2 marks**

Case 8 Collapse on the ward

A · 18 marks

Q1 Define cardiac arrest and name this rhythm. **2 marks**

MARKING SCHEME

- Clinical diagnosis: **loss of carotid/apical pulse + loss of consciousness** — 1
- **Ventricular fibrillation** — shockable rhythm — 1

Q2 Describe your immediate actions. **5 marks**

MARKING SCHEME

- Call for help / arrest team; note time — 1
- Start **CPR 30:2**, rate 100–120/min, depth 5–6 cm, full recoil, minimal interruption — 1
- **Immediate unsynchronised DC shock** — 200 J biphasic / 360 J monophasic — 1
- Resume CPR immediately for 2 min, then rhythm check; repeat shocks — 1
- Airway (bag-mask → advanced airway; then continuous compressions + 10 breaths/min), IV/IO access — 1

Q3 Which drugs, at what doses and when? **3 marks**

MARKING SCHEME

- **Adrenaline 1 mg IV** after the 2nd shock (lecture) / 3rd (ERC), then every 3–5 min — 1
- **Amiodarone 300 mg IV** after 3rd shock, further 150 mg after 5th — 1
- Lidocaine 1 mg/kg if amiodarone unavailable; MgSO4 2 g if torsades — 1

Q4 List the reversible causes you must look for. **3 marks**

MARKING SCHEME (5 H + 5 T; 3 MARKS IF ≥8 CORRECT)

- Hypoxia
- Hypovolaemia
- Hypothermia
- Hypo/hyperkalaemia
- Hydrogen ion (acidosis)
- Thrombosis — coronary
- Thrombosis — pulmonary
- Tamponade
- Tension pneumothorax
- Toxins

Q5 How would management differ if the monitor showed a normal-looking sinus rhythm but there was still no pulse? **3 marks**

MARKING SCHEME

- **PEA** — non-shockable: **no DC shock** — 1
- CPR + **adrenaline 1 mg IV as soon as possible**, then every 3–5 min — 1
- Aggressively treat 5H/5T (e.g. tamponade → pericardiocentesis, tension → decompression) — 1

Q6 Compare BLS and ACLS. **2 marks**

MARKING SCHEME

- **BLS**: out of hospital, by anyone; compressions 30:2 + AED; **no drugs** — 1
- **ACLS**: in hospital, by HCWs; advanced airway, manual defibrillation, **IV drugs**, treat causes, post-ROSC care — 1

DRUGS IN THIS CASE

Drug	Class	Key point
Adrenaline (epinephrine)	Sympathomimetic — α + β agonist (vasopressor)	1 mg IV every 3–5 min
Amiodarone	Class III antiarrhythmic	300 mg then 150 mg in VF/pVT
Lidocaine	Class Ib antiarrhythmic (Na-channel blocker)	Alternative to amiodarone
Magnesium sulphate	Electrolyte / membrane stabiliser	Torsade de pointes

Case 9 A swollen knee, then a swollen wrist — and 20 years later

A · 24 marks

A 7-year-old girl has a painful swollen right knee; a few days earlier her left wrist was swollen and recovered completely. She has had repeated tonsillitis over the past year.

Temp 38.5 °C | HR 110 | Right knee red, hot, tender | Hb 11 g/dL | ESR 64 | CRP +ve | ASOT 600 Todd | ECG: PR 0.24 s | Echo: normal

Q1 Give 4 differential diagnoses of arthritis in this child. **2 marks**

Q2 Apply the modified Jones criteria. Can you diagnose ARF? **4 marks**

Q3 Outline treatment with doses. **5 marks**

Q4 What secondary prophylaxis does she need and for how long? **3 marks**

Q5 Twenty years later she is 27 and pregnant, with dyspnoea, haemoptysis and palpitations. Exam: malar flush, tapping apex, loud S1, opening snap and a low-pitched mid-diastolic rumble with presystolic accentuation at the apex. Diagnosis, and why has she deteriorated now? **4 marks**

Q6 Mention complications of MS and the treatment options. **6 marks**

Case 9 A swollen knee, then a swollen wrist — and 20 years later

A · 24 marks

Q1 Give 4 differential diagnoses of arthritis in this child. **2 marks**

MARKING SCHEME (½ EACH)

- Rheumatic fever
- Juvenile idiopathic / rheumatoid arthritis
- Reactive arthritis (Shigella, Salmonella, Yersinia)
- Septic/gonococcal arthritis
- Sickle cell crisis
- Leukaemia, SLE, serum sickness

Q2 Apply the modified Jones criteria. Can you diagnose ARF? **4 marks**

MARKING SCHEME

- Major: **migratory polyarthritis** (1 major) — 1
- Minor: **fever**, **raised ESR/CRP**, **prolonged PR** (arthralgia not counted as arthritis is present) — 1
- Evidence of strep: **ASOT 600 (>333 Todd)** — 1
- 1 major + ≥2 minor + evidence of GAS infection → **acute rheumatic fever** — 1

Q3 Outline treatment with doses. **5 marks**

MARKING SCHEME

- Bed rest — 1
- Eradication: **benzathine penicillin G 1.2 MU IM single** (600,000 U if <27 kg) or oral penicillin V × 10 days (erythromycin if allergic) — 1
- **Aspirin 100 mg/kg/day in 4 doses × 3–5 days → 75 mg/kg/day × 4–8 weeks** — 1
- Steroids (prednisolone 1–2 mg/kg/day) only for severe carditis with cardiomegaly/HF — not needed here — 1
- Chorea: rest, sedatives (haloperidol/valproate); HF: diuretics, digoxin — 1

Q4 What secondary prophylaxis does she need and for how long? **3 marks**

MARKING SCHEME

- **Benzathine penicillin 1.2 MU IM every 3–4 weeks** (best agent) — 1
- No carditis (her case) → **at least 5 years or until age 21** (whichever longer) — 1
- Carditis without residual disease ≥10 y/till 25; with residual valve disease ≥10 y/till 40, sometimes lifelong — 1

Q5 Twenty years later she is 27 and pregnant, with dyspnoea, haemoptysis and palpitations. Exam: malar flush, tapping apex, loud S1, opening snap and a low-pitched mid-diastolic rumble with presystolic accentuation at the apex. Diagnosis, and why has she deteriorated now? **4 marks**

MARKING SCHEME

- **Rheumatic mitral stenosis** — 1
- Murmur heard best at apex, left lateral position, with the bell — 1
- Pregnancy ↑ cardiac output & HR → ↓ diastolic filling time → ↑ LA pressure → pulmonary congestion — 1
- Severity signs: **short S2–OS interval, long rumble**, signs of pulmonary HTN (loud P2, RV heave) — 1

Q6 Mention complications of MS and the treatment options. **6 marks**

MARKING SCHEME

- Complications: **AF**, **systemic embolism** (stroke), pulmonary oedema/HTN, RHF, haemoptysis, IE — 2
- Medical: diuretics, **β-blocker** (slows HR → longer filling), anticoagulation if AF/embolism/LA thrombus — 1
- Intervention: **percutaneous balloon mitral valvotomy** (if suitable valve, no LA thrombus, no significant MR) — can be done in pregnancy — 2
- Surgical: valvotomy or MVR (mechanical/bioprosthetic) — 1

KEY CRITERIA · MODIFIED JONES (ACNES + FIFA)

Major	Minor	Rule
Arthritis (75%, migratory large joints) · Carditis (50%) · subcutaneous Nodules (1%) · Erythema marginatum (3%) · Sydenham chorea (10%)	Fever · ↑ESR/CRP · First-degree heart block (long PR) · Arthralgia	**2 major or 1 major + 2 minor + evidence of GAS** (ASOT >333, rising titre, throat culture). Exceptions: chorea alone; recurrence in endemic areas

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Benzathine penicillin G	Long-acting β-lactam antibiotic (penicillin)	Eradication + secondary prophylaxis
Erythromycin	Macrolide antibiotic	Penicillin allergy
Aspirin (high dose)	NSAID / salicylate (anti-inflammatory dose)	Dramatic response in RF arthritis
Prednisolone	Corticosteroid	Severe carditis only
Haloperidol	Typical antipsychotic (D2 blocker)	Chorea

Case 10 Fever after a tooth extraction

A · 22 marks

A 28-year-old man with rheumatic heart disease since childhood (took long-acting penicillin for a couple of years only) had a tooth extracted a month ago. Since then he has intermittent fever and fatigue, treated with antipyretics. Today he developed sudden weakness of the left arm and leg.

Temp 38 °C | BP 125/75 | Pallor | Splinter haemorrhages | Tender nodules on finger pulps | Apical diastolic thrill + pansystolic murmur | Spleen palpable | Urine: microscopic haematuria

Q1 What is the diagnosis? Mention the peripheral signs you look for. **4 marks**

Q2 What is the most likely organism? Name the organisms in two other settings. **3 marks**

Q3 What are the two most important investigations? Describe how to do the first one. **5 marks**

Q4 How would you treat him? **4 marks**

Q5 Give 3 indications for surgery. **3 marks**

Q6 How could this have been prevented? **3 marks**

Case 10 Fever after a tooth extraction

A · 22 marks

Q1 What is the diagnosis? Mention the peripheral signs you look for. **4 marks**

MARKING SCHEME

- **Subacute infective endocarditis** on rheumatic mitral valve, complicated by **embolic stroke** (right hemisphere) — 1
- **Osler nodes** — tender nodules on fingers/toes (painful!) — 1
- **Janeway lesions** (painless palmar macules), **splinter haemorrhages**, **Roth spots**, conjunctival/palatal petechiae — 1
- Clubbing, splenomegaly, pallor, haematuria (immune-complex GN) — 1

Q2 What is the most likely organism? Name the organisms in two other settings. **3 marks**

MARKING SCHEME

- Post-dental, native valve → ***Streptococcus viridans*** — 1
- IV drug users → ***Staph aureus***, right-sided (tricuspid) — 1
- Early prosthetic valve (<1 y) → ***Staph epidermidis*** — 1

Q3 What are the two most important investigations? Describe how to do the first one. **5 marks**

MARKING SCHEME

- **Blood cultures**: **3 sets** from **3 different sites**, **≥1 h apart** — 1
- Strict **aseptic technique**, **before antibiotics** — the only test that guides therapy — 1
- **Echocardiography**: TTE first; **TEE more sensitive** (vegetation, abscess, prosthetic valves) — 1
- Others: CBC (anaemia, ↑TLC), ESR/CRP, urine, renal function — 1
- **CT brain** (infarct vs haemorrhage — affects anticoagulation/surgery) — 1

Q4 How would you treat him? **4 marks**

MARKING SCHEME

- Empirical IV bactericidal therapy after cultures: **ampicillin 12 g/day + (flu)cloxacillin 12 g/day + gentamicin 3 mg/kg/day** (or vancomycin + gentamicin if allergic) — 1
- Then by culture & sensitivity: e.g. penicillin G or ceftriaxone for strep viridans — 1
- Duration **4–6 weeks** IV (6 weeks prosthetic) — 1
- Monitor fever, CRP, renal function, gentamicin levels, repeat echo — 1

Q5 Give 3 indications for surgery. **3 marks**

MARKING SCHEME

- **Heart failure** (valve destruction / severe regurgitation) — 1
- **Uncontrolled infection** (abscess, fungal/resistant organism, persistent bacteraemia) — 1
- **Prevention of embolism** (large vegetation >10 mm with embolic event — as here) — 1

Q6 How could this have been prevented? **3 marks**

MARKING SCHEME

- Continue RF secondary prophylaxis as recommended (penicillin) — 1
- IE prophylaxis for high-risk patients before dental procedures: **amoxicillin 2 g orally 30–60 min before** — 1
- Penicillin allergy: azithromycin 500 mg (or clindamycin 600 mg); good dental hygiene — 1

KEY CRITERIA · MODIFIED DUKE CRITERIA (FOR DIAGNOSIS OF IE)

Major	Minor	Definite IE
1. Positive blood cultures (typical organism in 2 separate sets, or persistent bacteraemia) 2. Evidence of endocardial involvement on echo (vegetation, abscess, new dehiscence) or new regurgitation	Predisposition (heart lesion, IVDU) · fever ≥38 °C · vascular phenomena (emboli, Janeway, mycotic aneurysm) · immunological (Osler, Roth, GN, RF) · microbiology not meeting major	**2 major**, or **1 major + 3 minor**, or **5 minor**

Case 11 Right-sided chest pain after a long bus journey

A · 26 marks

A 38-year-old woman on the combined oral contraceptive pill presents with sudden severe right-sided chest pain that worsens on inspiration and breathlessness, after a 12-hour bus journey. Her left calf is swollen and tender.

BP 100/60 | HR 110 | RR 28 | SpO2 90% air | Temp 37.4 | Chest clear | ECG: sinus tachycardia, S1Q3T3 | CXR: near normal

Q1 Most probable diagnosis and the risk factors here? **3 marks**

Q2 Calculate her Wells score and interpret. **3 marks**

Q3 List the investigations and the findings you expect. **5 marks**

Q4 She is haemodynamically stable. Give her anticoagulation plan with doses and duration. **5 marks**

Q5 Two hours later: BP 75/45, raised JVP, syncope. Management? **5 marks**

Q6 When is an IVC filter indicated, and how do you prevent VTE in hospital patients? **3 marks**

Q7 When would you screen her for thrombophilia? **2 marks**

Case 11 Right-sided chest pain after a long bus journey

A · 26 marks

Q1 Most probable diagnosis and the risk factors here? **3 marks**

MARKING SCHEME

- **Acute pulmonary embolism** secondary to left leg DVT — 1
- Risk factors: **OCP** (oestrogen), **prolonged immobility/travel** — 1
- Pain is **non-cardiac (pleuritic)** — not angina — 1

Q2 Calculate her Wells score and interpret. **3 marks**

MARKING SCHEME

- Clinical DVT signs 3 + PE most likely 3 + HR >100 1.5 = **7.5** — 1
- >4 → **PE likely** (high probability) — 1
- → go straight to **CTPA**; D-dimer is not useful (only to exclude in low/intermediate probability) — 1

Q3 List the investigations and the findings you expect. **5 marks**

MARKING SCHEME

- **CTPA = first-line imaging** — filling defect in pulmonary arteries — 1
- **ABG**: hypoxaemia + hypocapnia (respiratory alkalosis) — 1
- **ECG**: sinus tachycardia (commonest), S1Q3T3, RBBB, RAD, AF — 1
- **Echo**: RV dilatation (RV:LV >1), D-shaped LV, **McConnell sign**; troponin/BNP for risk — 1
- Lower-limb venous Doppler; V/Q scan if contrast allergy/renal failure; CBC, coagulation baseline — 1

Q4 She is haemodynamically stable. Give her anticoagulation plan with doses and duration. **5 marks**

MARKING SCHEME

- Start anticoagulation **immediately** while awaiting imaging (high suspicion) — 1
- **Enoxaparin 1 mg/kg SC every 12 h** (or UFH 80 U/kg bolus → 18 U/kg/h, aPTT 1.5–2.5× control) — 1
- **Warfarin** started day 1–2 (5 mg), overlap ≥5 days until **INR 2–3** for 24 h — 1
- Or DOAC: rivaroxaban 15 mg bd × 3 wk → 20 mg daily (or apixaban 10 mg bd × 7 d → 5 mg bd) — 1
- Duration **3 months** (provoked) — **stop the OCP**; non-oestrogen contraception — 1

Q5 Two hours later: BP 75/45, raised JVP, syncope. Management? **5 marks**

MARKING SCHEME

- **Massive (high-risk) PE** with obstructive shock — 1
- O2 100%, careful fluids (≤500 mL — RV overload), **norepinephrine / dobutamine** — 1
- Bedside **echo** if CT not immediately possible (don't transfer unstable patients) — 1
- **Thrombolysis: alteplase 100 mg IV over 2 h** (or streptokinase 250,000 U over 30 min → 100,000 U/h for 12–24 h); UFH (not LMWH) around lysis — 1
- If lysis contraindicated/fails → **catheter or surgical embolectomy** — 1

Q6 When is an IVC filter indicated, and how do you prevent VTE in hospital patients? **3 marks**

MARKING SCHEME

- IVC filter: **anticoagulation contraindicated** or **recurrent PE despite adequate anticoagulation** — 1
- Prophylaxis: **enoxaparin 40 mg SC daily** (or UFH 5000 U SC bd–tds) in at-risk inpatients — 1
- Compression stockings, early mobilisation, hydration — 1

Q7 When would you screen her for thrombophilia? **2 marks**

MARKING SCHEME

- Unprovoked VTE <50 y, family history, recurrent/unusual site VTE, recurrent miscarriage — 1
- Her VTE was **provoked** (OCP + travel) → screening not routinely needed; test off anticoagulants — 1

KEY CRITERIA · WELLS SCORE FOR PE

Item	Points	Item	Points
Clinical signs of DVT	3	Previous DVT/PE	1.5
PE is the most likely diagnosis	3	Haemoptysis	1
HR >100	1.5	Active cancer	1
Immobilisation ≥3 d / surgery <4 wk	1.5	**>4 = PE likely → CTPA · ≤4 = unlikely → D-dimer**	

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Enoxaparin	Anticoagulant — LMWH (anti-Xa)	Preferred if stable; safe in pregnancy
Unfractionated heparin	Anticoagulant — indirect thrombin/Xa inhibitor	Unstable, renal failure, before lysis; HIT
Warfarin	Vitamin K antagonist	Teratogenic; many interactions
Rivaroxaban / apixaban	DOAC — direct factor Xa inhibitors	No INR monitoring
Alteplase / streptokinase	Fibrinolytics (thrombolytics)	Massive PE with shock
Norepinephrine / dobutamine	Vasopressor (α-agonist) / inotrope (β1-agonist)	Support in obstructive shock

Case 12 Chest pain that eases when he leans forward

B · 18 marks

A 30-year-old man had a flu-like illness 10 days ago. He now has sharp central chest pain, worse on lying flat and deep inspiration, relieved by sitting forward.

Temp 37.9 °C | BP 125/80 | Scratchy sound at left sternal edge, best leaning forward in expiration | ECG: concave ST elevation in I, II, aVF, V2–V6; PR depression; ST depression in aVR | Troponin: normal

Q1 Diagnosis and most likely cause? — **2 marks**

Q2 How does his ECG differ from STEMI? — **3 marks**

Q3 Treatment with doses? — **4 marks**

Q4 A week later he is breathless and dizzy. BP 80/60, falling 15 mmHg on inspiration, JVP very high, heart sounds muffled. Diagnosis and key bedside signs? — **4 marks**

Q5 How is it confirmed and treated? — **3 marks**

Q6 Name 2 features distinguishing chronic constrictive pericarditis. — **2 marks**

Case 12 Chest pain that eases when he leans forward

B · 18 marks

Q1 Diagnosis and most likely cause? **2 marks**

MARKING SCHEME
- **Acute pericarditis** — pericardial **friction rub is the hallmark** 1
- **Viral** (Coxsackie B, echovirus); others: TB, uraemia, autoimmune, post-MI (Dressler), radiation 1

Q2 How does his ECG differ from STEMI? **3 marks**

MARKING SCHEME
- ST elevation is **diffuse** (not in one territory) and **concave** (saddle-shaped) 1
- **PR depression** (specific), ST↓/PR↑ in aVR 1
- **No reciprocal changes, no Q waves** 1

Q3 Treatment with doses? **4 marks**

MARKING SCHEME
- Treat the cause; rest (avoid exercise until symptoms/CRP settle) 1
- **NSAID**: ibuprofen 600 mg tds or aspirin 750–1000 mg tds × 1–2 weeks, then taper; gastro-protection 1
- **Colchicine 0.5 mg bd (0.5 mg daily if <70 kg) × 3 months** — ↓ recurrence 1
- Steroids only if NSAIDs contraindicated/fail; high-dose aspirin post-MI 1

Q4 A week later he is breathless and dizzy. BP 80/60, falling 15 mmHg on inspiration, JVP very high, heart sounds muffled. Diagnosis and key bedside signs? **4 marks**

MARKING SCHEME
- **Cardiac tamponade** 1
- **Beck's triad**: hypotension + raised JVP + muffled heart sounds 1
- **Pulsus paradoxus** (SBP ↓ >10 mmHg on inspiration) 1
- JVP: preserved x, absent y descent; ECG low voltage + **electrical alternans**; CXR water-bottle heart 1

Q5 How is it confirmed and treated? **3 marks**

MARKING SCHEME
- **Echocardiography** (diagnostic): effusion with RA/RV diastolic collapse 1
- Urgent **pericardiocentesis** (echo-guided subxiphoid) / surgical drainage, pericardial window 1
- IV fluids as temporary support; **avoid diuretics & vasodilators**; send fluid (cells, AFB, cytology) 1

Q6 Name 2 features distinguishing chronic constrictive pericarditis. **2 marks**

MARKING SCHEME (1 EACH, ANY 2)
- **Kussmaul sign** (JVP rises on inspiration); **pericardial knock**; deep y descent
- Ascites before oedema; calcified thick pericardium on CT/MRI (best); dip-and-plateau trace → pericardiectomy

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Ibuprofen	NSAID (non-selective COX inhibitor)	Give with PPI
Aspirin (high dose)	Salicylate NSAID	Preferred post-MI pericarditis
Colchicine	Anti-inflammatory — microtubule (tubulin) inhibitor	Diarrhoea; ↓ recurrence
Prednisolone	Corticosteroid	↑ recurrence risk if used first-line

2 Chest cases

8 cases · 43 questions | COPD · asthma · CAP · TB · effusion · pneumothorax · IPF · lung cancer

Case 13 The smoker who can't finish a sentence

A · 26 marks

A 65-year-old man, 45 pack-years, has had a productive cough for most days of the last 3 years and progressive exertional dyspnoea. For 3 days his breathlessness has worsened and his sputum has become more and green. He is drowsy but rousable.

RR 30 | HR 112 | BP 135/85 | Temp 37.9 | Central cyanosis | Flapping tremor | Barrel chest, pursed lips, accessory muscles | Bilateral wheeze | Ankle oedema | ABG (air): pH 7.28 · PaCO2 68 · PaO2 48 · HCO3 31

Q1 What is the diagnosis? — **3 marks**

Q2 Interpret the ABG. — **3 marks**

Q3 Give his immediate management with doses. — **6 marks**

Q4 What other conditions must you exclude in an exacerbation, and how? — **3 marks**

Q5 After recovery, how do you confirm COPD and grade airflow limitation? — **3 marks**

Q6 Outline his long-term (stable) management. — **5 marks**

Q7 Compare "blue bloater" and "pink puffer". — **3 marks**

Case 13 The smoker who can't finish a sentence

A · 26 marks

Q1 What is the diagnosis? **3 marks**

MARKING SCHEME

- **Acute exacerbation of COPD** (↑dyspnoea + ↑sputum volume + ↑purulence = all 3 cardinal symptoms) 1
- Probably infective (fever, purulent sputum) 1
- Complicated by **acute-on-chronic type II respiratory failure** and cor pulmonale 1

Q2 Interpret the ABG. **3 marks**

MARKING SCHEME

- **Type II respiratory failure**: PaO2 <60 + PaCO2 >45 1
- **Respiratory acidosis** (pH 7.28) 1
- ↑HCO3 = partial metabolic compensation → **acute-on-chronic** hypercapnia 1

Q3 Give his immediate management with doses. **6 marks**

MARKING SCHEME

- **Controlled O2** by Venturi mask 24–28% → target **SpO2 88–92%**; repeat ABG in 30–60 min 1
- Nebulised **salbutamol 2.5–5 mg** + **ipratropium 0.5 mg** every 4–6 h (air-driven) 1
- **Prednisolone 40 mg orally × 5 days** (or hydrocortisone 100 mg IV) 1
- **Antibiotic** (purulent sputum): amoxicillin–clavulanate 1 g bd or doxycycline 100 mg bd, 5–7 days 1
- **NIV (BiPAP)** — pH <7.35 and PaCO2 >45 despite medical therapy 1
- Intubation if NIV fails/↓GCS/pH <7.25 worsening; DVT prophylaxis, diuretic for oedema 1

Q4 What other conditions must you exclude in an exacerbation, and how? **3 marks**

MARKING SCHEME (ANY 6, ½ EACH)

- Pneumonia — CXR, CRP/procalcitonin
- Pneumothorax — CXR/US
- Pleural effusion — CXR/US
- PE — D-dimer, CTPA
- Pulmonary oedema — ECG, echo, BNP
- Arrhythmia (AF) — ECG

Q5 After recovery, how do you confirm COPD and grade airflow limitation? **3 marks**

MARKING SCHEME

- **Spirometry** is required: **post-bronchodilator FEV1/FVC <0.70** 1
- GOLD grade by FEV1 %pred: 1 ≥80 · 2 50–79 · 3 30–49 · 4 <30 1
- Plus CXR (hyperinflation, bullae), CBC (polycythaemia), α1-antitrypsin if young/basal emphysema, echo (cor pulmonale) 1

Q6 Outline his long-term (stable) management. **5 marks**

MARKING SCHEME

- **Smoking cessation** — only measure that slows decline (NRT, varenicline) 1
- **Influenza + pneumococcal vaccines**; pulmonary rehabilitation 1
- **LAMA + LABA** (e.g. tiotropium 18 µg daily + formoterol/indacaterol) + SABA PRN 1
- Add ICS if frequent exacerbations and eosinophils ≥300; roflumilast/azithromycin selected 1
- **LTOT ≥15 h/day** if PaO2 ≤55 mmHg (or 55–60 with cor pulmonale/polycythaemia) when stable 1

Q7 Compare "blue bloater" and "pink puffer". **3 marks**

MARKING SCHEME

- Blue bloater (chronic bronchitis): productive cough, cyanosis, hypercapnia, ↓drive, **cor pulmonale/oedema**, unmatched V/Q 1½
- Pink puffer (emphysema): dyspnoea, little sputum, barrel chest, pursed lips, thin, adequate O2 via hyperventilation, matched V/Q 1½

KEY CRITERIA · RESPIRATORY FAILURE & NIV

Type I	Type II	NIV indicated	NIV contraindicated
PaO2 <60, PaCO2 normal/low (pneumonia, oedema, PE, ARDS)	PaO2 <60 + PaCO2 >45 (COPD, asthma, NM, chest wall, overdose)	pH <7.35 & PaCO2 >45 after initial therapy; alert, airway protected	Coma, vomiting, untreated pneumothorax, facial trauma, haemodynamic instability

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Salbutamol	SABA — short-acting β2-agonist	Tremor, tachycardia, $\downarrow K^+$
Ipratropium	SAMA — short-acting muscarinic antagonist	Dry mouth, urinary retention
Tiotropium	LAMA — long-acting muscarinic antagonist	Once daily
Formoterol / salmeterol / indacaterol	LABA — long-acting β2-agonists	Never alone in asthma
Budesonide / fluticasone	Inhaled corticosteroids (ICS)	Pneumonia risk in COPD; oral thrush
Prednisolone	Systemic corticosteroid	40 mg × 5 days
Amoxicillin–clavulanate	Aminopenicillin + β-lactamase inhibitor	Covers H. influenzae, Moraxella
Doxycycline	Tetracycline antibiotic	Avoid in pregnancy/children
Roflumilast	PDE4 inhibitor	Weight loss, diarrhoea
Varenicline	Nicotinic receptor partial agonist	Smoking cessation

Case 14 The quiet chest

A · 22 marks

A 22-year-old woman with known asthma (uses salbutamol inhaler 4–5 times daily, not on preventer) presents with a 2-day history of worsening wheeze after a cold. She speaks in single words.

RR 32 | HR 125 | BP 110/70 | SpO2 91% air | PEF 30% of best | Very quiet breath sounds

ABG: PaO2 56 · PaCO2 44 · pH 7.37

Q1 Grade the severity and justify. **3 marks**

Q2 Give immediate treatment with doses. **7 marks**

Q3 Which investigations and monitoring? **3 marks**

Q4 Outside the acute attack, how is asthma diagnosed objectively? **3 marks**

Q5 What are the criteria for discharge, and what long-term treatment does she need? **4 marks**

Q6 Give 4 differential diagnoses of wheeze. **2 marks**

Case 14 The quiet chest

A · 22 marks

Q1 Grade the severity and justify. **3 marks**

MARKING SCHEME

- **Life-threatening acute asthma** 1
- PEF **<33%**, SpO2 **<92%**, **silent chest** 1
- **"Normal" PaCO2 (44)** in a breathless asthmatic = tiring → near-fatal danger sign 1

Q2 Give immediate treatment with doses. **7 marks**

MARKING SCHEME

- Sit up; **O2** high-flow to SpO2 **94–98%** 1
- **Salbutamol 5 mg nebulised** (O2-driven), repeat every 15–30 min / back-to-back 1
- **Ipratropium 0.5 mg nebulised** every 4–6 h 1
- **Hydrocortisone 100 mg IV every 6 h** (or prednisolone 40–50 mg orally) 1
- **MgSO4 1.2–2 g IV over 20 min** (life-threatening / poor response) 1
- Consider IV aminophylline 5 mg/kg load over 20 min (not if on theophylline) / IV salbutamol 1
- Urgent **ICU review** — intubation if exhaustion, ↓GCS, rising PaCO2; CXR to exclude pneumothorax; **no sedatives** 1

Q3 Which investigations and monitoring? **3 marks**

MARKING SCHEME

- **PEF** before and 15–30 min after each treatment; continuous SpO2 1
- **ABG** (repeat within 1 h); serum **K^+** (β2-agonists + steroids ↓K^+), glucose 1
- **CXR** (pneumothorax, consolidation); CBC, CRP if infection suspected 1

Q4 Outside the acute attack, how is asthma diagnosed objectively? **3 marks**

MARKING SCHEME

- Spirometry: obstruction (FEV1/FVC <0.7) with **reversibility: FEV1 ↑ >12% (and 200 mL)** after SABA 1
- **PEF ↑ ≥15%** 15–20 min after SABA, or diurnal PEF variability (serial monitoring) 1
- Methacholine (bronchial provocation) test; FeNO, eosinophils, allergy tests (atopy/IgE) 1

Q5 What are the criteria for discharge, and what long-term treatment does she need? **4 marks**

MARKING SCHEME

- Discharge when **PEF >75%** best, stable on inhalers ≥12–24 h 1
- Oral prednisolone to complete ≥5 days 1
- Start **inhaled corticosteroid** controller (e.g. low–medium-dose ICS–formoterol; SABA overuse = uncontrolled) 1
- Check inhaler technique, written action plan, avoid triggers, follow-up within 48 h–1 week 1

Q6 Give 4 differential diagnoses of wheeze. **2 marks**

MARKING SCHEME (½ EACH)

- COPD
- Cardiac asthma (pulmonary oedema)
- Foreign body / upper airway obstruction
- Bronchiectasis / CF / bronchiolitis / anaphylaxis

KEY CRITERIA · SEVERITY OF ACUTE ASTHMA

Moderate	Acute severe (any one)	Life-threatening (any one)	Near-fatal
PEF 50–75%, no severe features	PEF 33–50% · RR ≥25 · HR ≥110 · can't complete sentences	PEF <33% · SpO2 <92% / PaO2 <60 · normal PaCO2 · silent chest · cyanosis · poor effort · exhaustion · arrhythmia · hypotension · confusion	Raised PaCO2 or needing mechanical ventilation

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Salbutamol	SABA — β2-agonist	Reliever; ↓K^+
Ipratropium	SAMA — antimuscarinic	Add-on in severe attack
Hydrocortisone / prednisolone	Systemic corticosteroids	Effect takes 4–6 h — give early
Magnesium sulphate	Bronchial smooth-muscle relaxant (Ca-channel effect)	Single dose IV
Aminophylline	Methylxanthine (PDE inhibitor)	Narrow therapeutic index; arrhythmia, seizures
Budesonide–formoterol	ICS + LABA combination	Maintenance & reliever (GINA)
Montelukast	Leukotriene-receptor antagonist	Add-on controller

Case 15 Rusty sputum and a pleuritic stitch

A · 22 marks

A 62-year-old smoker with hypertension and diabetes has had a productive cough of thick yellow sputum for 4 days with fever, chills and malaise. Yesterday he developed right chest pain worse on inspiration. He is confused about the date.

BP 152/90 | HR 112 | RR 31 | Temp 38.5 | SpO2 91% | Right mammary/axillary: dull, ↑TVF, bronchial breathing, crepitations | WBC 17,000 (neutrophils 70%) | Urea 9 mmol/L

Q1 Diagnosis? Which signs indicate consolidation? — **3 marks**

Q2 Calculate CURB-65 and decide the site of care. — **3 marks**

Q3 Which investigations? — **4 marks**

Q4 Give the antibiotic regimen and supportive care. — **5 marks**

Q5 A 77-year-old COPD patient has dry cough, diarrhoea, headache, confusion, low-grade fever and Na^+ 126. CXR: right lower interstitial shadowing; chest exam normal. Likely organism and how do you confirm and treat it? — **3 marks**

Q6 Fever persists on day 5 and the right base is stony dull. What has happened and what do you do? — **4 marks**

Case 15 Rusty sputum and a pleuritic stitch

A · 22 marks

Q1 Diagnosis? Which signs indicate consolidation? **3 marks**

MARKING SCHEME

- **Community-acquired pneumonia** (right middle/lower lobe), likely *S. pneumoniae* — 1
- Consolidation: ↓movement, ↑**TVF**, **dull percussion**, **bronchial breathing** — 1
- ↑vocal resonance, whispering pectoriloquy, egophony, crepitations ± rub — 1

Q2 Calculate CURB-65 and decide the site of care. **3 marks**

MARKING SCHEME

- Confusion 1 + Urea >7 1 + RR ≥30 1 + BP 0 + age <65 0 = **3** — 1
- ≥3 = **severe CAP** — 1
- → **hospital admission, consider ICU/HDU** — 1

Q3 Which investigations? **4 marks**

MARKING SCHEME

- **CXR** (diagnosis, multilobar, effusion, abscess) / lung US (dynamic air bronchogram) — 1
- CBC, CRP/procalcitonin, renal (urea), electrolytes, LFT, glucose, **ABG** — 1
- **Blood cultures ×2** + **sputum Gram stain & culture** before antibiotics — 1
- **Urinary antigens** (pneumococcus, Legionella), PCR panel/SARS-CoV-2, HIV test if risk — 1

Q4 Give the antibiotic regimen and supportive care. **5 marks**

MARKING SCHEME

- Severe CAP: **β-lactam IV** (ceftriaxone 1–2 g IV daily, or amoxicillin–clavulanate 1.2 g IV tds) — 1
- **+ macrolide** (clarithromycin 500 mg IV bd / azithromycin 500 mg daily) — covers atypicals — 1
- Or respiratory quinolone (levofloxacin 750 mg / moxifloxacin 400 mg daily) — 1
- Duration 5–7 days; switch to oral when afebrile 48 h & stable — 1
- O2 (SpO2 94–98%), IV fluids, antipyretic/analgesia, chest physiotherapy, glucose control, DVT prophylaxis — 1

Q5 A 77-year-old COPD patient has dry cough, diarrhoea, headache, confusion, low-grade fever and Na^+ 126. CXR: right lower interstitial shadowing; chest exam normal. Likely organism and how do you confirm and treat it? **3 marks**

MARKING SCHEME

- ***Legionella pneumophila*** (atypical: GI symptoms, hyponatraemia, confusion) — 1
- **Urinary Legionella antigen** (or PCR) — 1
- **Macrolide** (azithromycin) or **quinolone** (levofloxacin); NIV if needed — 1

Q6 Fever persists on day 5 and the right base is stony dull. What has happened and what do you do? **4 marks**

MARKING SCHEME

- Parapneumonic effusion / **empyema** (other causes of non-resolution: abscess, resistant/atypical organism, TB, obstruction by tumour, drug fever) — 1
- **US-guided diagnostic tap**: pH, glucose, LDH, Gram stain, culture — 1
- Pus / pH <7.2 / glucose low / organisms → **wide-bore chest drain** ± intrapleural fibrinolytics — 1
- Antibiotics for **2–4 weeks** (+ anaerobic cover); thoracoscopy/decortication if loculated — 1

KEY CRITERIA · CURB-65

C	U	R	B	65	0–1	2	≥3
Confusion	Urea >7 mmol/L	RR ≥30	SBP <90 or DBP ≤60	Age ≥65	Home	Hospital	Severe → consider ICU

DRUGS IN THIS CASE

Drug	Class	Key point
Ceftriaxone	3rd-generation cephalosporin (β-lactam)	Pneumococcus, H. influenzae
Amoxicillin–clavulanate	Penicillin + β-lactamase inhibitor	Oral switch option
Clarithromycin / azithromycin	Macrolides	Atypicals; QT prolongation
Levofloxacin / moxifloxacin	Respiratory fluoroquinolones	Tendinopathy, QT; avoid if TB possible
Enoxaparin 40 mg	LMWH (prophylactic dose)	DVT prophylaxis

Case 16 A month of cough that antibiotics didn't touch

A · 22 marks

A 23-year-old man has had a cough for one month that did not respond to a course of amoxicillin. He reports night sweats, loss of appetite and weight loss over 2–3 months. His uncle, who lives with him, "coughs blood".

Temp 37.8 (evening) | Weight 54 kg | Fine crepitations right interscapular area

CXR: right upper-lobe cavity with patchy infiltrates

Q1 What is the most likely diagnosis? What is the first investigation? **3 marks**

Q2 How do you confirm the diagnosis, and what other tests? **4 marks**

Q3 Prescribe treatment with doses for 54 kg and duration. **6 marks**

Q4 Name the key side effect of each first-line drug. **4 marks**

Q5 How would you monitor response, and what if he remains smear-positive at 2 months? **3 marks**

Q6 What will you do for the household? **2 marks**

Case 16 A month of cough that antibiotics didn't touch

A · 22 marks

Q1 What is the most likely diagnosis? What is the first investigation? **3 marks**

MARKING SCHEME

- **Post-primary (reactivation) pulmonary TB**, cavitary, upper lobe — 1
- **Sputum for AFB (Ziehl-Neelsen) — 3 consecutive (early-morning) samples** — 1
- + **Xpert MTB/RIF** (result <2 h, detects rifampicin resistance — WHO initial test) — 1

Q2 How do you confirm the diagnosis, and what other tests? **4 marks**

MARKING SCHEME

- **Culture (Löwenstein-Jensen) = gold standard** — up to 6–12 weeks; culture even smear-negative samples — 1
- **Drug-susceptibility testing** on the first isolate — 1
- **HIV test** (all TB patients), LFT/renal/uric acid baseline, visual acuity & colour vision — 1
- TST/IGRA show infection only (not disease); CXR alone never confirms TB — 1

Q3 Prescribe treatment with doses for 54 kg and duration. **6 marks**

MARKING SCHEME

- **Intensive phase 2 months HRZE** — 1
- **Isoniazid 5 mg/kg** ≈ 300 mg daily (max 300) — 1
- **Rifampicin 10 mg/kg** ≈ 600 mg daily (max 600) — 1
- **Pyrazinamide 25 mg/kg** ≈ 1.25–1.5 g daily · **ethambutol 15 mg/kg** ≈ 800 mg daily — 1
- **Continuation phase 4 months HR** (total 6 months), fixed-dose combinations, **DOT** — 1
- **Pyridoxine 10–25 mg daily** to prevent INH neuropathy — 1

Q4 Name the key side effect of each first-line drug. **4 marks**

MARKING SCHEME (1 EACH)

- Isoniazid — hepatitis, **peripheral neuropathy**
- Rifampicin — hepatitis, **orange urine/tears**, **enzyme induction** (OCP failure, warfarin)
- Pyrazinamide — hepatitis, **hyperuricaemia/gout**
- Ethambutol — **optic neuritis** (red-green colour vision)

Q5 How would you monitor response, and what if he remains smear-positive at 2 months? **3 marks**

MARKING SCHEME

- Sputum smear at end of month 2, 5 and 6; weight, symptoms, LFT if symptoms — 1
- Still positive → check adherence, repeat culture + DST (Xpert) for resistance — 1
- MDR-TB (H + R resistant) → second-line regimen (levofloxacin/moxifloxacin, bedaquiline, linezolid…) by TB centre — 1

Q6 What will you do for the household? **2 marks**

MARKING SCHEME

- Notify; **contact tracing**: symptom screen, CXR, TST/IGRA for household contacts — 1
- Treat active TB; latent TB → preventive therapy (isoniazid 6–9 months or 3HP); isolate smear-positive patient ≥2 weeks of therapy — 1

DRUGS IN THIS CASE

Drug	Class	Key point
Isoniazid (H)	First-line anti-TB — mycolic acid synthesis inhibitor	Give pyridoxine
Rifampicin (R)	First-line anti-TB — rifamycin (RNA polymerase inhibitor)	Potent enzyme inducer
Pyrazinamide (Z)	First-line anti-TB	Gout
Ethambutol (E)	First-line anti-TB (arabinosyl transferase inhibitor)	Eye checks
Pyridoxine	Vitamin B6	Prevents INH neuropathy
Levofloxacin / moxifloxacin	Fluoroquinolones — 2nd-line anti-TB	MDR-TB
Bedaquiline	Diarylquinoline (ATP-synthase inhibitor)	MDR-TB; QT

Case 17 A stony-dull right base

B · 16 marks

A 35-year-old woman has 6 weeks of low-grade fever, dry cough, right-sided pleuritic pain and increasing dyspnoea.

Right base: ↓movement, stony dull, ↓TVF, ↓breath sounds | Trachea shifted left | CXR: right effusion up to mid-zone

Pleural fluid: protein 5.2 g/dL (serum 7) · LDH 420 (serum 300) · glucose 60 · cells 90% lymphocytes · ADA ↑

Q1 Is this transudate or exudate? Apply the criteria. **3 marks**

Q2 What is the most likely cause, and what is your DD? **4 marks**

Q3 How do you confirm the diagnosis? **3 marks**

Q4 Main lines of treatment? **3 marks**

Q5 Give the indications for pleurodesis. **3 marks**

Case 17 A stony-dull right base

B · 16 marks

Q1 Is this transudate or exudate? Apply the criteria. **3 marks**

MARKING SCHEME

- Fluid/serum protein 5.2/7 = 0.74 (>0.5) — 1
- Fluid/serum LDH 420/300 = 1.4 (>0.6) — 1
- → **Exudate** (Light's criteria — any one is enough) — 1

Q2 What is the most likely cause, and what is your DD? **4 marks**

MARKING SCHEME

- **Tuberculous pleural effusion** — lymphocytic exudate, high ADA, subacute course — 1
- Malignancy (haemorrhagic, cytology) — 1
- Parapneumonic/empyema (neutrophils, organisms), PE/infarction — 1
- Collagen disease (rheumatoid — very low glucose; SLE). Transudates: HF, cirrhosis, nephrotic — 1

Q3 How do you confirm the diagnosis? **3 marks**

MARKING SCHEME

- **Pleural biopsy** (thoracoscopic/closed) → caseating granuloma + culture — highest yield — 1
- Fluid AFB smear/culture/Xpert (low yield), ADA — 1
- Sputum AFB, CXR after drainage (underlying lung), TST/IGRA, HIV test — 1

Q4 Main lines of treatment? **3 marks**

MARKING SCHEME

- Standard **anti-TB therapy 2 HRZE + 4 HR** — 1
- **Therapeutic aspiration** for symptom relief (≤1.5 L at a time — re-expansion oedema) — 1
- Follow up for residual pleural thickening; treat underlying lung TB — 1

Q5 Give the indications for pleurodesis. **3 marks**

MARKING SCHEME

- Rapidly re-accumulating (e.g. malignant) effusion — 1
- Recurrent, bilateral or secondary pneumothorax — 1
- Persistent air leak / high-risk occupations (divers, pilots) — 1

KEY CRITERIA · LIGHT'S CRITERIA & PLEURAL-FLUID PATTERNS

Exudate if ANY	TB	Parapneumonic / empyema	Malignant	Rheumatoid	Transudate
Protein F/S >0.5 · LDH F/S >0.6 · fluid LDH >⅔ ULN serum	Lymphocytes, ↑ADA	Neutrophils, ↓pH <7.2, ↓glucose, organisms/pus	Blood-stained, cytology +	Very ↓glucose	HF, cirrhosis, nephrotic; clear, low protein

DRUGS IN THIS CASE

Drug	Class	Key point
Isoniazid, rifampicin, pyrazinamide, ethambutol	First-line anti-tuberculous drugs	6-month regimen
Talc (pleurodesis)	Sclerosing agent	Obliterates pleural space

Case 18 Sudden breathlessness in a tall young smoker

B · 16 marks

A 20-year-old tall thin male smoker develops sudden left-sided chest pain and breathlessness while sitting at his desk.

RR 24 | HR 105 | BP 125/80 | SpO2 95% | Left: ↓movement, hyper-resonant, absent breath sounds & TVF | Trachea central

Q1 Diagnosis and classification? **3 marks**

Q2 Investigation, and how is size judged? **2 marks**

Q3 Manage if (a) rim 1 cm and comfortable, (b) rim 3 cm and breathless. **4 marks**

Q4 His BP falls to 80/50, trachea deviates to the right, neck veins distend. What is happening and what do you do immediately? **4 marks**

Q5 A 68-year-old COPD patient has a 1.5 cm pneumothorax. How does his management differ? **3 marks**

Case 18 Sudden breathlessness in a tall young smoker

B · 16 marks

Q1 Diagnosis and classification? **3 marks**

MARKING SCHEME

- **Left spontaneous pneumothorax** — 1
- **Primary** (no underlying lung disease; tall thin smoker, apical blebs) — 1
- Others: secondary (COPD, TB, asthma), traumatic, iatrogenic; closed/open/tension — 1

Q2 Investigation, and how is size judged? **2 marks**

MARKING SCHEME

- **Erect CXR** (visible pleural line, no lung markings) — or chest US — 1
- Large = rim ≥2 cm at hilum level (BTS); CT if complex — 1

Q3 Manage if (a) rim 1 cm and comfortable, (b) rim 3 cm and breathless. **4 marks**

MARKING SCHEME

- (a) Small, asymptomatic → **observation** ± high-flow O2 (speeds resorption), review/CXR in 2–4 days — 1
- (b) Large/symptomatic → **needle aspiration** (16–18 G, 2nd ICS MCL) — 1
- Aspiration fails → **intercostal chest drain** (safe triangle) with underwater seal — 1
- **Stop smoking**; avoid flying until resolved, never dive unless pleurodesis — 1

Q4 His BP falls to 80/50, trachea deviates to the right, neck veins distend. What is happening and what do you do immediately? **4 marks**

MARKING SCHEME

- **Tension pneumothorax** (one-way valve → mediastinal shift → obstructive shock) — 1
- Clinical diagnosis — **do NOT wait for CXR** — 1
- **Immediate needle decompression**: 2nd ICS mid-clavicular line (or 4th–5th ICS anterior axillary line) — 1
- Followed by **chest drain**; O2 — 1

Q5 A 68-year-old COPD patient has a 1.5 cm pneumothorax. How does his management differ? **3 marks**

MARKING SCHEME

- **Secondary** pneumothorax — poorly tolerated — 1
- **Admit**; 1–2 cm → aspiration; >2 cm or breathless → **chest drain** — 1
- Persistent leak / recurrence → thoracoscopic surgery or pleurodesis; controlled O2 88–92% — 1

DRUGS IN THIS CASE

Drug	Class	Key point
High-flow oxygen	Medical gas	↑ resorption of pleural air ×4 (not in CO2 retainers)
Lidocaine 1%	Local anaesthetic (amide, Na-channel blocker)	For aspiration / drain insertion
Talc / tetracycline	Pleural sclerosants	Chemical pleurodesis

Case 19 Dry cough, clubbing and velcro crackles

A · 16 marks

A 66-year-old retired glass worker, ex-smoker, has 18 months of progressive exertional dyspnoea and a persistent dry cough.

Finger clubbing | Fine bibasal end-inspiratory "velcro" crackles | SpO2 94% rest → 86% on walking

FEV1 68% · FVC 62% · FEV1/FVC 0.86 · DLCO 45%

Q1 Interpret the lung function. **3 marks**

Q2 Most likely diagnosis and differentials? **4 marks**

Q3 What investigations? **4 marks**

Q4 Main lines of management of IPF. **5 marks**

Case 19 Dry cough, clubbing and velcro crackles

A · 16 marks

Q1 Interpret the lung function. **3 marks**

MARKING SCHEME

- **Restrictive pattern**: ↓FVC with normal/↑FEV1/FVC (>0.7) — 1
- ↓**DLCO** → impaired gas transfer (parenchymal disease) — 1
- Exercise desaturation → significant ILD — 1

Q2 Most likely diagnosis and differentials? **4 marks**

MARKING SCHEME

- **Interstitial lung disease** — likely **idiopathic pulmonary fibrosis (UIP)**, the commonest ILD — 1
- Occupational: **silicosis** (glass worker), asbestosis — 1
- CTD-ILD (RA, scleroderma), hypersensitivity pneumonitis (birds), sarcoidosis — 1
- Drug-induced: amiodarone, methotrexate, nitrofurantoin, bleomycin — 1

Q3 What investigations? **4 marks**

MARKING SCHEME

- **HRCT chest** — UIP: basal subpleural reticulation, **honeycombing**, traction bronchiectasis — 1
- Full PFT + 6-minute walk test; ABG — 1
- Serology: ANA, RF, anti-CCP, ANCA, ENA (exclude CTD) — 1
- BAL / transbronchial or surgical lung biopsy if HRCT not typical — MDT decision; echo for pulmonary HTN — 1

Q4 Main lines of management of IPF. **5 marks**

MARKING SCHEME

- **Antifibrotics: pirfenidone or nintedanib** (slow FVC decline) — 1
- Ambulatory/long-term **O2**; pulmonary rehabilitation — 1
- Vaccines (influenza, pneumococcal); treat GERD; stop exposures — 1
- **Steroids/immunosuppression are NOT used in IPF** (used in CTD-ILD, HP, sarcoid) — 1
- Refer for **lung transplantation**; palliative care for breathlessness (opioids) — 1

DRUGS IN THIS CASE

Drug	Class	Key point
Pirfenidone	Antifibrotic (TGF-β pathway modulator)	Photosensitivity, GI upset
Nintedanib	Antifibrotic — tyrosine-kinase inhibitor	Diarrhoea, ↑LFT
Prednisolone	Corticosteroid	Not for IPF

Case 20 Blood-streaked sputum and a hoarse voice

C · 16 marks

A 70-year-old man, smoker for 50 years, has had a persistent cough for 2 months with blood-tinged sputum, weight loss and, recently, a hoarse voice. His face and neck look swollen in the mornings.

Clubbing | Dilated veins over upper chest | Na^+ 124 mmol/L (euvolaemic) | CXR: right hilar mass

Q1 Most likely diagnosis and the features of local invasion here? **4 marks**

Q2 Explain the hyponatraemia and name 2 other paraneoplastic syndromes. **3 marks**

Q3 Outline the diagnostic and staging work-up. **5 marks**

Q4 Main lines of treatment and referral. **4 marks**

Case 20 Blood-streaked sputum and a hoarse voice

C · 16 marks

Q1 Most likely diagnosis and the features of local invasion here? **4 marks**

MARKING SCHEME

- **Bronchogenic carcinoma** (central — squamous or small cell) 1
- Hoarseness → **recurrent laryngeal nerve** invasion (usually left) 1
- Facial swelling + dilated chest veins → **superior vena cava obstruction** 1
- Others: phrenic palsy, dysphagia, effusion, Horner/Pancoast 1

Q2 Explain the hyponatraemia and name 2 other paraneoplastic syndromes. **3 marks**

MARKING SCHEME

- **SIADH** (ectopic ADH) — typical of **small-cell carcinoma** 1
- Hypercalcaemia (PTHrP) — squamous cell 1
- Ectopic ACTH/Cushing, Lambert-Eaton (SCLC), clubbing/HPOA 1

Q3 Outline the diagnostic and staging work-up. **5 marks**

MARKING SCHEME

- **Contrast CT chest** (+ upper abdomen) 1
- Tissue: **fibreoptic bronchoscopy + biopsy** (central lesion) 1
- Peripheral lesions → US/CT-guided biopsy; sputum cytology 1
- Mediastinal nodes: EBUS / mediastinoscopy 1
- Staging: **PET-CT**, CT/MRI brain, LFT, Ca^{2+} → TNM, MDT 1

Q4 Main lines of treatment and referral. **4 marks**

MARKING SCHEME

- **Refer to oncology/MDT** (level C topic) 1
- NSCLC early stage → surgery (lobectomy) ± chemo; stage III → chemoradiation 1
- Stage IV → chemotherapy, **targeted therapy** (EGFR/ALK), immunotherapy; SCLC → chemo-radiotherapy 1
- SVCO: dexamethasone, **SVC stent**, radiotherapy; SIADH: fluid restriction; smoking cessation 1

DRUGS IN THIS CASE

Drug	Class	Key point
Dexamethasone	Corticosteroid	Oedema in SVCO / brain mets
Cisplatin + etoposide	Platinum alkylating-like + topoisomerase II inhibitor	SCLC chemotherapy
Osimertinib / erlotinib	EGFR tyrosine-kinase inhibitors	EGFR-mutant adenocarcinoma
Pembrolizumab	Immune checkpoint (PD-1) inhibitor	Immune-related adverse effects

3 Tropical cases

11 cases · 63 questions | Liver abscess · dysentery · brucella · meningitis · hepatitis · mono · FUO · bilharzia · malaria · sepsis · HIV

Case 21 Right upper-quadrant pain after dysentery

A · 22 marks

A 35-year-old farmer from rural Egypt had bloody mucoid diarrhoea 2 months ago that settled without treatment. For 10 days he has had fever with rigors, night sweats and right upper-quadrant pain referred to the right shoulder, with a dry cough and hiccups.

Temp 39 °C | Tender hepatomegaly, intercostal tenderness | Dull right lung base | WBC 16,000, no eosinophilia | ESR 70 | ALP mildly ↑ | Stool: no cysts | US: single 7 cm hypoechoic round lesion, right lobe

Q1 What is the most likely diagnosis? Give 2 differentials. **3 marks**

Q2 Why is the stool negative, and what is the mainstay of diagnosis? **3 marks**

Q3 Give the treatment with doses. **5 marks**

Q4 When would you aspirate/drain? **4 marks**

Q5 What does the aspirate look like? **2 marks**

Q6 List 5 complications. **3 marks**

Q7 How would a hydatid cyst differ in investigations and treatment? **2 marks**

Case 21 Right upper-quadrant pain after dysentery

A · 22 marks

Q1 What is the most likely diagnosis? Give 2 differentials. **3 marks**

MARKING SCHEME

- **Amoebic liver abscess** (*E. histolytica*) — most common extra-intestinal amoebiasis **1**
- Pyogenic liver abscess (*E. coli*, *Klebsiella*) **1**
- Infected hydatid cyst, HCC/necrotic tumour **1**

Q2 Why is the stool negative, and what is the mainstay of diagnosis? **3 marks**

MARKING SCHEME

- Abscess appears 1–3 months after dysentery; **stool is usually negative** (+ in only ~50%) **1**
- **Serology** (IHA / ELISA antibodies) — positive in ~99% after 1 week **1**
- Plus imaging (US, CT); stool antigen/PCR; LAMP assay **1**

Q3 Give the treatment with doses. **5 marks**

MARKING SCHEME

- Start treatment **before** serology results **1**
- Tissue amoebicide: **metronidazole 750–800 mg orally tds × 10 days** (IV if vomiting) **1**
- Alternative: tinidazole 2 g/day × 3 days (or nitazoxanide) **1**
- **Then luminal amoebicide**: paromomycin 25 mg/kg/day in 3 doses × 10 days **or** diloxanide furoate 500 mg tds × 10 days **1**
- Expect defervescence within 3–4 days; abscess ≤10 cm usually cured medically **1**

Q4 When would you aspirate/drain? **4 marks**

MARKING SCHEME

- **Uncertain diagnosis** (exclude pyogenic) **1**
- **No response after 3–5 days** of metronidazole **1**
- **Left-lobe abscess >10 cm** (risk of pericardial rupture) / impending rupture **1**
- Rupture into pleura (empyema) → surgical drainage **1**

Q5 What does the aspirate look like? **2 marks**

MARKING SCHEME

- **"Anchovy sauce"** — thick, chocolate-brown, **odourless**, acellular debris **1**
- Sterile on culture; trophozoites only in the **wall** of the abscess **1**

Q6 List 5 complications. **3 marks**

MARKING SCHEME (ANY 5)

- Rupture into peritoneum (peritonitis)
- Into pleura (empyema) / lung (hepatobronchial fistula)
- Into pericardium (tamponade)
- Secondary bacterial infection
- Brain abscess (dissemination)
- IVC obstruction / Budd-Chiari

Q7 How would a hydatid cyst differ in investigations and treatment? **2 marks**

MARKING SCHEME

- Dog contact, **eosinophilia**, US daughter cysts/**water-lily sign**, calcification; never aspirate blindly (anaphylaxis) **1**
- **Albendazole 400 mg bd** + PAIR or surgery for complex/large cysts **1**

Case 22 Thirty stools a day

A · 20 marks

(a) A 42-year-old man has 3 days of diarrhoea with mucus and blood, abdominal cramps, tenesmus and fever 38.7 °C. A week of an antibiotic from his GP did not help. **(b)** In a refugee camp, a 25-year-old woman develops profuse painless watery "rice-water" stools and vomiting; she is drowsy with sunken eyes, BP 80/50.

Q1 (a) Define dysentery and give the differential diagnosis. **3 marks**

Q2 (a) Which investigations? **3 marks**

Q3 (a) Stool culture grows Shigella. Treatment and complications? **4 marks**

Q4 (a) If microscopy had shown amoebic trophozoites, how would you treat? **2 marks**

Q5 (b) What is the diagnosis and how is it confirmed? **3 marks**

Q6 (b) Manage her. **5 marks**

Case 22 Thirty stools a day

A · 20 marks

Q1 (a) Define dysentery and give the differential diagnosis. **3 marks**

MARKING SCHEME

- Dysentery = diarrhoea with **blood and mucus** + tenesmus/fever (invasive colitis) 1
- Bacterial: **Shigella**, Campylobacter, EHEC, Salmonella; C. difficile (after antibiotics) 1
- Parasitic: **amoebic dysentery**, schistosomiasis; non-infectious: IBD, ischaemic colitis 1

Q2 (a) Which investigations? **3 marks**

MARKING SCHEME

- **Fresh stool microscopy** — trophozoites with ingested RBCs (amoeba), pus cells, ova 1
- **Stool culture + sensitivity**; stool antigen/PCR; C. difficile toxin if recent antibiotics 1
- CBC, electrolytes (Na^+), renal; sigmoidoscopy ± biopsy (flask ulcers, IBD) 1

Q3 (a) Stool culture grows Shigella. Treatment and complications? **4 marks**

MARKING SCHEME

- **Rehydration** (ORS / IV) + nutrition; avoid antimotility drugs 1
- Antibiotic per sensitivity: **ciprofloxacin 500 mg bd × 3–5 days** or **azithromycin 500 mg daily × 3 days** (ampicillin/TMP-SMX if sensitive) 1
- Intestinal complications: toxic megacolon, perforation, protein-losing enteropathy 1
- Extra-intestinal: **HUS**, **reactive arthritis (Reiter)**, hyponatraemia + seizures 1

Q4 (a) If microscopy had shown amoebic trophozoites, how would you treat? **2 marks**

MARKING SCHEME

- **Metronidazole 750 mg tds × 5–10 days** (tinidazole/nitazoxanide) 1
- **Followed by luminal agent** (diloxanide furoate 500 mg tds or paromomycin × 10 days) 1

Q5 (b) What is the diagnosis and how is it confirmed? **3 marks**

MARKING SCHEME

- **Cholera** with severe dehydration / hypovolaemic shock 1
- In epidemics: clinical/epidemiological; dark-field microscopy (darting vibrios) 1
- Culture on **TCBS agar**, PCR; ↑haematocrit/serum protein (haemoconcentration), electrolytes, glucose, renal 1

Q6 (b) Manage her. **5 marks**

MARKING SCHEME

- **Rapid IV rehydration** — Ringer lactate 100 mL/kg (30 mL/kg in first 30 min, rest over 2½ h) 1
- Then **ORS** to match ongoing losses until diarrhoea stops 1
- Correct K^+, hypoglycaemia, acidosis; monitor urine output 1
- Antibiotic shortens illness: **doxycycline 300 mg single dose** (or azithromycin 1 g, tetracycline, co-trimoxazole) 1
- Notify, isolate, safe water & sanitation, oral cholera vaccine for camp 1

Case 23 The cheese-maker with back pain and night sweats

A · 20 marks

A 40-year-old man who makes white cheese from raw goat's milk at home has had 5 weeks of evening fever that comes and goes, drenching night sweats with a strange moldy smell, arthralgia, low-back pain and a painful swollen right testis.

Temp 38.6 | Mild hepatosplenomegaly | Tender lower lumbar spine | Swollen tender right testis/epididymis | WBC 3,800 (relative lymphocytosis) | ALT mildly ↑

Q1 Most likely diagnosis and source? — **3 marks**

Q2 How do you confirm the diagnosis? — **4 marks**

Q3 Give the treatment with doses. How does spondylitis change it? — **5 marks**

Q4 List 4 other complications of brucellosis. — **2 marks**

Q5 What is the Brucellin skin test's value? — **2 marks**

Q6 Define chronic brucellosis and give prevention measures. — **4 marks**

Case 23 The cheese-maker with back pain and night sweats

A · 20 marks

Q1 Most likely diagnosis and source? **3 marks**

MARKING SCHEME

- **Acute/subacute brucellosis** (*B. melitensis*) 1
- Source: **unpasteurised milk / cheese** from goats (also contact, aerosols, lab workers) 1
- Focal complications: **spondylitis** + **epididymo-orchitis** 1

Q2 How do you confirm the diagnosis? **4 marks**

MARKING SCHEME

- **Blood culture** (prolonged incubation; warn lab — biohazard); **bone-marrow culture** more sensitive = gold standard 1
- **Serum agglutination test (SAT)**: ≥1:160 in endemic areas (≥1:80 non-endemic) 1
- or a **4-fold rise** between samples ≥2 weeks apart 1
- ELISA IgM/IgG, PCR; MRI spine for spondylitis; US testis 1

Q3 Give the treatment with doses. How does spondylitis change it? **5 marks**

MARKING SCHEME

- Principles: **combination**, **intracellular-active** drugs, **prolonged** 1
- **Doxycycline 100 mg bd × 6 weeks + rifampicin 600–900 mg daily × 6 weeks** 1
- or doxycycline × 6 wk + **streptomycin 1 g IM daily × 2–3 weeks** (or gentamicin 5 mg/kg × 7 d) 1
- **Spondylitis**: doxycycline + aminoglycoside-containing regimen, total **≥3 months** 1
- Pregnancy/children <8: rifampicin + co-trimoxazole 1

Q4 List 4 other complications of brucellosis. **2 marks**

MARKING SCHEME (½ EACH)

- **Endocarditis** (main cause of death)
- Neurobrucellosis (meningoencephalitis)
- Sacroiliitis, peripheral arthritis
- Abortion in pregnancy, pancytopenia, hepatitis

Q5 What is the Brucellin skin test's value? **2 marks**

MARKING SCHEME

- Delayed hypersensitivity; induration ≥6 mm at 24 h — positive only in **chronic** brucellosis 1
- Negative in acute disease; a negative test excludes brucellosis 1

Q6 Define chronic brucellosis and give prevention measures. **4 marks**

MARKING SCHEME

- Symptoms >1 year after diagnosis 1
- Localised infection (spondylitis, abscess, uveitis) or relapse with objective evidence 1
- **Pasteurise/boil milk**, avoid raw dairy 1
- Animal vaccination & testing, protective equipment for farmers/vets, lab precautions 1

Case 24 Fever, headache and a rash that doesn't blanch

A · 24 marks

A 19-year-old university student living in a crowded dormitory presents with 12 hours of fever, severe headache, vomiting and photophobia. He is now drowsy.

Temp 39.6 | BP 95/60 | HR 120 | GCS 13 | Neck stiffness, positive Kernig | Petechial non-blanching rash on legs | No papilloedema, no focal signs

Q1 What is the most likely diagnosis and organism? **2 marks**

Q2 What is your sequence of immediate actions? **4 marks**

Q3 Give the empirical regimen with doses. **5 marks**

Q4 Describe the expected CSF and compare with viral and TB meningitis. **5 marks**

Q5 What do you do for his roommates? **3 marks**

Q6 A 45-year-old diabetic has 3 weeks of headache, low fever, drowsiness and a left 6th nerve palsy. CSF: lymphocytes, protein 250, glucose very low. Diagnosis, complications and treatment? **5 marks**

Case 24 Fever, headache and a rash that doesn't blanch

A · 24 marks

Q1 What is the most likely diagnosis and organism? **2 marks**

MARKING SCHEME

- **Acute bacterial meningitis with meningococcal septicaemia** 1
- *Neisseria meningitidis* (Gram −ve diplococci); petechial/purpuric rash, crowding 1

Q2 What is your sequence of immediate actions? **4 marks**

MARKING SCHEME

- ABC, O2, IV access, **blood cultures** (+ PCR), glucose, CBC, coagulation, lactate 1
- **Antibiotics immediately** — do not delay for LP or CT 1
- **LP** if no contraindication (CT first if focal signs, papilloedema, seizures, GCS ≤10, immunocompromised) 1
- Here: shock + purpura → resuscitate first, LP can be deferred; fluids for septic shock 1

Q3 Give the empirical regimen with doses. **5 marks**

MARKING SCHEME

- **Ceftriaxone 2 g IV every 12 h** (or cefotaxime 2 g q4–6h) 1
- **+ vancomycin 15–20 mg/kg IV q8–12h** (penicillin-resistant pneumococcus) 1
- **+ ampicillin 2 g IV q4h** if age >50, pregnancy, alcoholism, immunosuppression (Listeria) 1
- **Dexamethasone 0.15 mg/kg (10 mg) IV q6h × 4 days**, first dose before/with antibiotics 1
- Duration: meningococcus 7 d, pneumococcus 10–14 d, Listeria 21 d 1

Q4 Describe the expected CSF and compare with viral and TB meningitis. **5 marks**

MARKING SCHEME

- Bacterial: **turbid**, ↑↑ **neutrophils** 1
- ↑↑ protein, ↓↓ **glucose** (CSF/serum <0.4); Gram stain, culture, latex antigen, PCR 1
- Viral: clear, lymphocytes, normal glucose, normal/mildly ↑ protein 1
- TB: clear/cobweb, lymphocytes, ↑↑ protein, ↓ glucose, ↑ADA, ZN/GeneXpert 1
- Normal CSF: clear, <5 lymphocytes, protein 15–45 mg/dL, glucose ≥60% of blood 1

Q5 What do you do for his roommates? **3 marks**

MARKING SCHEME

- **Notify** public health; identify close contacts 1
- Chemoprophylaxis: **rifampicin 600 mg bd × 2 days** or **ciprofloxacin 500 mg single dose** or **ceftriaxone 250 mg IM single** (pregnancy) 1
- Meningococcal vaccine (ACWY/B) for contacts/outbreak 1

Q6 A 45-year-old diabetic has 3 weeks of headache, low fever, drowsiness and a left 6th nerve palsy. CSF: lymphocytes, protein 250, glucose very low. Diagnosis, complications and treatment? **5 marks**

MARKING SCHEME

- **Tuberculous meningitis** (subacute) 1
- Basal adhesions → **cranial nerve palsies** (2, 3, 4, 6, 7), **hydrocephalus** 1
- Vasculitis → strokes; seizures; tuberculoma; SIADH 1
- **2 HRZE + 7–10 HR** = **9–12 months** 1
- + **dexamethasone** (tapering 6–8 weeks); VP shunt for hydrocephalus 1

KEY CRITERIA · CSF PATTERNS

	Normal	Bacterial	Viral	TB
Appearance	Clear	Turbid	Clear	Clear / fibrin web
Cells	<5 lymphocytes	↑↑ neutrophils	↑ lymphocytes	↑ lymphocytes
Protein	15–45 mg/dL	↑↑	N / mild ↑	↑↑
Glucose	≥60% blood	↓↓	Normal	↓

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Ceftriaxone / cefotaxime	3rd-generation cephalosporins	Good CSF penetration
Vancomycin	Glycopeptide	Resistant pneumococcus
Ampicillin	Aminopenicillin	Listeria (cephalosporins don't cover it)
Dexamethasone	Corticosteroid	↓ deafness/neurological sequelae
Rifampicin / ciprofloxacin	Rifamycin / fluoroquinolone	Contact prophylaxis
Acyclovir	Antiviral — nucleoside analogue (DNA polymerase)	Add if HSV encephalitis suspected

Case 25 Yellow eyes — read the serology

A · 20 marks

A 26-year-old man has had 1 week of anorexia, nausea, malaise and fever, followed by dark urine and jaundice, after which the fever subsided. He had a tattoo 3 months ago.

Tender hepatomegaly | ALT 1450 · AST 1100 | ALP 180 | Bilirubin 7 mg/dL | INR 1.2 | HBsAg +ve · anti-HBc IgM +ve | Anti-HAV IgM −ve · anti-HCV −ve

Q1 What is the diagnosis? Interpret the serology. **3 marks**

Q2 Is it severe? What would indicate acute liver failure? **3 marks**

Q3 Outline management. **4 marks**

Q4 How do you protect his wife? **3 marks**

Q5 A pregnant woman in her third trimester has acute hepatitis after drinking well water during an outbreak. HAV and HBV markers are negative. What do you suspect and why does it matter? **3 marks**

Q6 What distinguishes HDV co-infection from superinfection? **2 marks**

Q7 What is the treatment of chronic HCV in Egypt? **2 marks**

Case 25 Yellow eyes — read the serology

A · 20 marks

Q1 What is the diagnosis? Interpret the serology. **3 marks**

MARKING SCHEME

- **Acute hepatitis B** — 1
- HBsAg = current infection; **anti-HBc IgM = acute (recent)** infection — 1
- Source: tattoo (parenteral); IP 30–180 days — 1

Q2 Is it severe? What would indicate acute liver failure? **3 marks**

MARKING SCHEME

- Not severe: **INR <1.5**, no encephalopathy — 1
- Acute liver failure = jaundice + **INR >1.5** + **encephalopathy** without prior liver disease — 1
- Fulminant: massive necrosis with ↓consciousness within 8 weeks of onset → transplant centre — 1

Q3 Outline management. **4 marks**

MARKING SCHEME

- Outpatient (alert, no coagulopathy); rest until jaundice fades; balanced diet — 1
- **Stop alcohol & hepatotoxic drugs**; IV fluids if vomiting — 1
- LFT + PT every 5–7 days × 2 weeks, then every 14 days until normal — 1
- Antiviral (**entecavir or tenofovir**) only if severe/fulminant; test HBsAg at 6 months (chronicity) — 1

Q4 How do you protect his wife? **3 marks**

MARKING SCHEME

- Test her HBsAg, anti-HBs, anti-HBc — 1
- If non-immune: **HBIG + HBV vaccine** (0, 1, 6 months) — 1
- Condoms until she is immune; don't share razors/toothbrushes — 1

Q5 A pregnant woman in her third trimester has acute hepatitis after drinking well water during an outbreak. HAV and HBV markers are negative. What do you suspect and why does it matter? **3 marks**

MARKING SCHEME

- **Hepatitis E** (faeco-oral, water-borne) — 1
- Confirm with **anti-HEV IgM** — 1
- **15–25% mortality in pregnancy** (fulminant failure) → admit, monitor INR/glucose, transplant if failure — 1

Q6 What distinguishes HDV co-infection from superinfection? **2 marks**

MARKING SCHEME

- Co-infection: HDV IgM + **anti-HBc IgM** — usually self-limited — 1
- Superinfection: HDV IgM + HBsAg with **anti-HBc IgG** (chronic carrier) — severe/fulminant or chronic — 1

Q7 What is the treatment of chronic HCV in Egypt? **2 marks**

MARKING SCHEME

- Easy-to-treat (bilirubin <1.2, albumin >3.5, platelets >150,000): **sofosbuvir + daclatasvir** — 1
- Difficult-to-treat: **sofosbuvir + daclatasvir + ribavirin**; HCC surveillance US + AFP every 6 months in cirrhosis — 1

KEY CRITERIA · HEPATITIS B SEROLOGY

HBsAg	Anti-HBc IgM	Anti-HBc IgG	Anti-HBs	Interpretation
+	+	–	–	**Acute HBV**
+	–	+	–	**Chronic HBV** (>6 months; HBeAg/DNA for activity)
–	–	+	+	Past infection, immune
–	–	–	+	Vaccinated
–	+	–	–	Window period

DRUGS IN THIS CASE

Drug	Class	Key point
Entecavir / tenofovir	Nucleos(t)ide analogues (HBV polymerase inhibitors)	Severe acute / chronic HBV
HBIG	Hepatitis B immunoglobulin (passive immunity)	Post-exposure, with vaccine
HBV vaccine	Recombinant HBsAg (inactivated subunit)	0, 1, 6 months
Sofosbuvir	DAA — NS5B polymerase inhibitor	HCV backbone
Daclatasvir	DAA — NS5A inhibitor	With sofosbuvir
Ribavirin	Nucleoside analogue antiviral	Haemolytic anaemia; teratogenic
Peg-interferon	Immunomodulator (cytokine)	HDV

Case 26 Sore throat and a rash after amoxicillin

A · 18 marks

A 19-year-old student has 10 days of fever, severe fatigue and sore throat. He was given amoxicillin and 2 days later developed a generalised maculopapular rash.

Temp 38.4 | Exudative tonsillitis, palatal petechiae | Bilateral posterior cervical lymphadenopathy | Spleen tip palpable | WBC 14,000, lymphocytes 65%, 25% atypical | ALT 160 | Platelets 110,000

Q1 Diagnosis and explanation of the rash? **3 marks**

Q2 How do you confirm it? **3 marks**

Q3 Give differential diagnoses. **3 marks**

Q4 Outline management and advice. **5 marks**

Q5 Mention 4 complications/associations of EBV. **2 marks**

Q6 A renal-transplant recipient 2 months post-op has fever, hepatitis and pneumonitis; Monospot negative. Likely cause and treatment? **2 marks**

Case 26 Sore throat and a rash after amoxicillin

A · 18 marks

Q1 Diagnosis and explanation of the rash? **3 marks**

MARKING SCHEME
- **Infectious mononucleosis** (EBV / HHV-4) 1
- Triad: fever + pharyngitis + lymphadenopathy (posterior cervical most suggestive) 1
- Rash = typical **ampicillin/amoxicillin reaction in EBV** (not true allergy) 1

Q2 How do you confirm it? **3 marks**

MARKING SCHEME
- CBC: lymphocytosis >50% with **>10% atypical lymphocytes**, mild thrombocytopenia, ↑ALT 1
- **Heterophile antibody test** (Monospot / Paul-Bunnell) 1
- EBV-specific: **VCA-IgM** (acute); EBNA appears later (past infection); PCR 1

Q3 Give differential diagnoses. **3 marks**

MARKING SCHEME (½ EACH)
- Streptococcal pharyngitis
- CMV mononucleosis (no pharyngitis/LN)
- Toxoplasmosis
- **Acute HIV** seroconversion
- HHV-6, viral hepatitis
- Acute leukaemia / lymphoma

Q4 Outline management and advice. **5 marks**

MARKING SCHEME
- Rest, fluids, paracetamol, saltwater gargles (self-limiting 6–8 weeks) 1
- **Stop amoxicillin**; avoid ampicillin-type drugs 1
- **Avoid contact sports for ~3–4 weeks** — risk of splenic rupture 1
- **Prednisolone 40–60 mg/day × 2–3 days then taper over 1–2 weeks** only for **airway obstruction** (tonsillar oedema), AIHA or severe thrombocytopenia 1
- No antivirals routinely; avoid alcohol while LFT abnormal 1

Q5 Mention 4 complications/associations of EBV. **2 marks**

MARKING SCHEME (½ EACH)
- Splenic rupture, airway obstruction
- Hepatitis, AIHA, thrombocytopenia
- Encephalitis, GBS, myocarditis
- Burkitt, Hodgkin, nasopharyngeal Ca, PTLD

Q6 A renal-transplant recipient 2 months post-op has fever, hepatitis and pneumonitis; Monospot negative. Likely cause and treatment? **2 marks**

MARKING SCHEME
- **CMV** infection (1–4 months post-transplant) — confirm with **CMV PCR viraemia** / IgM 1
- **IV ganciclovir** (valganciclovir orally for less severe); foscarnet if resistant 1

DRUGS IN THIS CASE

Drug	Class	Key point
Amoxicillin / ampicillin	Aminopenicillins	**Rash in EBV** — avoid
Paracetamol	Analgesic / antipyretic	Hepatotoxic in overdose
Prednisolone	Corticosteroid	Only for complications
Ganciclovir / valganciclovir	Nucleoside analogue antivirals	Myelosuppression
Foscarnet	Pyrophosphate analogue antiviral	Nephrotoxic

Case 27 Five weeks of fever, no answer yet

A · 16 marks

A 55-year-old man has had fever up to 38.8 °C for 5 weeks with weight loss. One week of inpatient investigation (CBC, urine, CXR, routine cultures) has not revealed a cause. He has taken several antibiotic courses.

Q1 Does he have FUO? State the definition. **3 marks**

Q2 Give the main categories of causes with percentages and examples. **4 marks**

Q3 Outline your stepwise approach. **5 marks**

Q4 Name the other 3 FUO categories. **2 marks**

Q5 Give 2 drugs that commonly cause drug fever. **2 marks**

Case 27 Five weeks of fever, no answer yet

A · 16 marks

Q1 Does he have FUO? State the definition. **3 marks**

MARKING SCHEME

- Temperature **≥38.3 °C** on several occasions — 1
- Duration **≥3 weeks** — 1
- No diagnosis after **1 week inpatient** (or 3 outpatient visits / 3 inpatient days) → yes, classic FUO — 1

Q2 Give the main categories of causes with percentages and examples. **4 marks**

MARKING SCHEME

- **Infection ~40%**: TB, abscess, IE, brucella, typhoid, UTI/prostatitis, osteomyelitis, CMV/EBV/HIV, malaria — 1
- **Malignancy ~25%**: lymphoma, leukaemia, **renal cell carcinoma**, HCC, metastases — 1
- **Autoimmune ~15%**: adult Still's, SLE, GCA/PMR, PAN, RA — 1
- **Miscellaneous ~10%** (drug fever, FMF, thyroiditis, sarcoid, factitious); **undiagnosed ~10%** — 1

Q3 Outline your stepwise approach. **5 marks**

MARKING SCHEME

- **Confirm fever** with a temperature chart; stop non-essential drugs (drug fever) — 1
- Repeat detailed history (travel, animals, raw milk, sex, occupation, drugs) & examination daily — 1
- Screen: CBC + film, ESR/CRP, LFT, **blood cultures ×3**, urine culture, CXR, Mantoux/IGRA, brucella & HIV serology — 1
- Second line: **CT chest/abdomen/pelvis**, echo (IE), ANA/RF/ANCA, protein electrophoresis, **bone marrow** — 1
- Third: PET-CT, tissue biopsy (LN, liver, temporal artery), endoscopy, laparoscopy; **avoid blind antibiotics/steroids** if stable — 1

Q4 Name the other 3 FUO categories. **2 marks**

MARKING SCHEME

- **Nosocomial** (C. difficile, drugs, PE, septic thrombophlebitis, sinusitis) · **neutropenic** (≤500; fungi, bacteria, herpes) — 1
- **HIV-associated** (>4 weeks OP / 3 days IP: MAC, CMV, PCP, lymphoma, drugs) — 1

Q5 Give 2 drugs that commonly cause drug fever. **2 marks**

MARKING SCHEME (1 EACH)

- β-lactam antibiotics, isoniazid, methyldopa, phenytoin, procainamide, quinidine

💊 DRUGS IN THIS CASE

Drug	Class	Key point
β-lactams (penicillins, cephalosporins)	Cell-wall synthesis inhibitors	Common cause of drug fever
Phenytoin	Antiepileptic (Na-channel blocker)	Drug fever, lymphadenopathy
Methyldopa	Central α2-agonist antihypertensive	Drug fever, hepatitis, haemolysis

Case 28 The farmer who vomited blood

A · 16 marks

A 45-year-old farmer from the Nile Delta, who has swum in canals since childhood, presents with haematemesis. He reports years of abdominal discomfort and occasional bloody diarrhoea.

BP 95/60 | HR 110 | Huge firm splenomegaly | Liver palpable, firm | No jaundice | Albumin 3.6 | Platelets 70,000 | US: periportal (pipe-stem) fibrosis, normal liver texture

Q1 Diagnosis? — **3 marks**

Q2 Immediate management of the bleed (main lines). — **4 marks**

Q3 How is schistosomiasis diagnosed? — **3 marks**

Q4 Specific treatment and prevention? — **3 marks**

Q5 Name the stages of the disease and 3 other system complications. — **3 marks**

Case 28 The farmer who vomited blood

A · 16 marks

Q1 Diagnosis? **3 marks**

MARKING SCHEME

- **Hepatosplenic (intestinal) schistosomiasis** — *S. mansoni* 1
- **Periportal fibrosis** → **presinusoidal portal hypertension** → bleeding oesophageal varices 1
- Hypersplenism (thrombocytopenia); liver function preserved until late 1

Q2 Immediate management of the bleed (main lines). **4 marks**

MARKING SCHEME

- Resuscitate: 2 large-bore IV lines, crystalloids, **crossmatch, restrictive transfusion (Hb 7–8)** 1
- **Vasoactive drug**: terlipressin 2 mg IV q4h or octreotide 50 µg bolus → 50 µg/h 1
- **Antibiotic prophylaxis** (ceftriaxone 1 g/day) 1
- **Urgent endoscopy** (<12 h) with **band ligation**; balloon tamponade/TIPS if uncontrolled 1

Q3 How is schistosomiasis diagnosed? **3 marks**

MARKING SCHEME

- **Stool (or urine) for ova** — lateral spine (mansoni) / terminal spine (haematobium) 1
- **Rectal snip / tissue biopsy**: eggs + granuloma 1
- Serology — can't separate past from active; useful in travellers; eosinophilia in acute phase 1

Q4 Specific treatment and prevention? **3 marks**

MARKING SCHEME

- **Praziquantel 40 mg/kg orally on one day** (single or 2 divided doses) 1
- Safe in pregnancy & lactation; doesn't reverse established fibrosis 1
- Prevention: avoid canal water, snail control, sanitation, mass treatment 1

Q5 Name the stages of the disease and 3 other system complications. **3 marks**

MARKING SCHEME

- Cercarial dermatitis → **Katayama fever** (2–8 wk: fever, urticaria, eosinophilia) → chronic 1
- Hepatic stages: hepatomegaly → HSM → huge spleen → ascites 1
- Cor pulmonale, myelopathy/encephalopathy, glomerulopathy; *haematobium*: haematuria, hydronephrosis, **bladder SCC** 1

Case 29 Back from Africa with fever and confusion

B · 18 marks

A 32-year-old engineer returned from Nigeria 10 days ago. He took no prophylaxis. He has 4 days of high fever with rigors, headache, vomiting and now confusion and dark urine.

Temp 40.1 | BP 90/60 | GCS 11 | Jaundice, splenomegaly | Hb 7.5 | Platelets 45,000 | Glucose 50 mg/dL | Creatinine 3.1 | Thin film: ring forms, 8% parasitaemia

Q1 Diagnosis? List the features of severity in this patient. **4 marks**

Q2 How is malaria diagnosed? **3 marks**

Q3 Main lines of treatment. **5 marks**

Q4 How would you treat *P. vivax* malaria, and what test is needed first? **3 marks**

Q5 How could this have been prevented? **3 marks**

Case 29 Back from Africa with fever and confusion

B · 18 marks

Q1 Diagnosis? List the features of severity in this patient. **4 marks**

MARKING SCHEME

- **Severe *Plasmodium falciparum* malaria** (malignant tertian) 1
- **Cerebral malaria** (↓GCS) 1
- Severe anaemia, **hypoglycaemia**, AKI, shock, jaundice 1
- Hyperparasitaemia (>5%), dark urine → **blackwater fever** (intravascular haemolysis) 1

Q2 How is malaria diagnosed? **3 marks**

MARKING SCHEME

- **Giemsa-stained thick film** (detect parasites) + **thin film** (species, % parasitaemia) 1
- Repeat films 12–24-hourly ×3 if negative 1
- Rapid antigen (dipstick) tests; PCR; serology useless acutely 1

Q3 Main lines of treatment. **5 marks**

MARKING SCHEME

- ICU; **IV artesunate** (standard of care for severe malaria) 1
- If unavailable: **IV quinine** (watch hypoglycaemia, QT) + doxycycline/clindamycin 1
- Complete with oral **ACT** (artemether–lumefantrine) once able to swallow 1
- **IV dextrose** for hypoglycaemia; transfusion for severe anaemia; fluids cautiously (ARDS risk) 1
- Dialysis for AKI; anticonvulsants for seizures; monitor parasitaemia 1

Q4 How would you treat *P. vivax* malaria, and what test is needed first? **3 marks**

MARKING SCHEME

- **Chloroquine** (ACT if chloroquine-resistant) for blood stage 1
- + **primaquine × 14 days** to eradicate liver **hypnozoites** (prevents relapse) 1
- Check **G6PD** first — primaquine causes haemolysis in deficiency 1

Q5 How could this have been prevented? **3 marks**

MARKING SCHEME

- Bite avoidance: **permethrin-treated bed nets**, repellents (DEET), long sleeves 1
- **Chemoprophylaxis**: atovaquone–proguanil, doxycycline or mefloquine 1
- Seek care for any fever within months of return; vaccines where indicated 1

Case 30 The catheterised patient who became confused

B · 18 marks

A 72-year-old diabetic woman has had a urinary catheter for 10 days after hip surgery. She becomes febrile, confused and breathless.

Temp 39.2 | BP 82/45 (MAP 57) | HR 125 | RR 26 | Cloudy urine | WBC 21,000 | Lactate 4.5 mmol/L | Creatinine 2.4 | Platelets 90,000

Q1 Calculate qSOFA and give the diagnosis. **3 marks**

Q2 Investigations? **3 marks**

Q3 Outline management (main lines). **6 marks**

Q4 Define sepsis and septic shock (Sepsis-3). **2 marks**

Q5 List 4 elements of the CAUTI prevention bundle. **2 marks**

Q6 Interpret procalcitonin values. **2 marks**

Case 30 The catheterised patient who became confused

B · 18 marks

Q1 Calculate qSOFA and give the diagnosis. **3 marks**

MARKING SCHEME

- qSOFA: altered mentation 1 + RR ≥22 1 + SBP ≤100 1 = **3** (≥2 = high risk) 1
- **Sepsis from catheter-associated UTI (CAUTI)** — hospital-acquired (>48 h) 1
- Likely **septic shock** if vasopressors needed for MAP ≥65 with lactate >2 after fluids 1

Q2 Investigations? **3 marks**

MARKING SCHEME

- **Blood cultures ×2 + urine culture (from new catheter) before antibiotics** 1
- CBC, lactate, renal, LFT, coagulation (DIC), glucose, ABG, procalcitonin 1
- CXR, renal US (obstruction/abscess) — imaging only if safe to transfer 1

Q3 Outline management (main lines). **6 marks**

MARKING SCHEME

- **Broad-spectrum IV antibiotics within the first hour** — e.g. piperacillin–tazobactam 4.5 g q6–8h or meropenem 1 g q8h (MDR risk), ± vancomycin/linezolid if MRSA risk 1
- **Crystalloid 30 mL/kg** (hypotension, lactate ≥4) 1
- **Norepinephrine** to maintain **MAP ≥65 mmHg** 1
- **Source control**: **remove/replace the catheter** within 6 h 1
- Hydrocortisone if refractory shock; PRBC if Hb <7; glucose control 1
- DVT & stress-ulcer prophylaxis; re-assess daily to **de-escalate**, 7–10 days total 1

Q4 Define sepsis and septic shock (Sepsis-3). **2 marks**

MARKING SCHEME

- Sepsis = life-threatening **organ dysfunction** from dysregulated host response to infection (SOFA ↑ ≥2) 1
- Septic shock = vasopressors for **MAP ≥65** + **lactate >2 mmol/L** despite adequate fluids 1

Q5 List 4 elements of the CAUTI prevention bundle. **2 marks**

MARKING SCHEME (½ EACH)

- Insert only with proper indication; consider alternatives
- Sterile insertion by trained staff, hand hygiene
- Closed system, bag below bladder, secure to thigh
- Ask daily "is it still needed?" — remove early

Q6 Interpret procalcitonin values. **2 marks**

MARKING SCHEME

- <0.5 ng/mL: sepsis unlikely · 0.5–2: possible 1
- 2–10: likely sepsis · falling PCT guides stopping antibiotics 1

KEY CRITERIA · QSOFA & SOFA

qSOFA (outside ICU, ≥2)	SOFA (ICU, ↑ ≥2) organ systems
Altered mentation · RR ≥22 · SBP ≤100	PaO2/FiO2 · platelets · bilirubin · MAP/vasopressors · creatinine/urine output · GCS

DRUGS IN THIS CASE

Drug	Class	Key point
Piperacillin–tazobactam	Antipseudomonal penicillin + β-lactamase inhibitor	Broad Gram −ve/anaerobes
Meropenem	Carbapenem	ESBL / MDR Gram −ve
Vancomycin / linezolid	Glycopeptide / oxazolidinone	MRSA cover
Norepinephrine	Vasopressor (α1-agonist)	First-line in septic shock
Hydrocortisone	Corticosteroid	Refractory shock (relative adrenal insufficiency)
Pantoprazole	Proton-pump inhibitor	Stress-ulcer prophylaxis

Case 31 The needle-stick, and the patient behind it

B · 16 marks

A nurse sustains a deep needle-stick injury while taking blood from a 34-year-old man admitted with 3 weeks of progressive dyspnoea, dry cough, fever, oral thrush and weight loss. His CXR shows bilateral perihilar ground-glass shadowing; SpO2 drops on walking.

Q1 What is the immediate first aid and post-exposure prophylaxis for the nurse? **4 marks**

Q2 What is the likely diagnosis in the patient, and how is HIV confirmed? **4 marks**

Q3 Define AIDS and give 4 opportunistic infections with their CD4 thresholds. **3 marks**

Q4 Main lines of management for the patient. **5 marks**

Case 31 The needle-stick, and the patient behind it

B · 16 marks

Q1 What is the immediate first aid and post-exposure prophylaxis for the nurse? **4 marks**

MARKING SCHEME

- Encourage bleeding, wash with soap and water, don't scrub/suck; report incident — 1
- Test source (with consent) and nurse baseline: HIV, HBV, HCV — 1
- **PEP within hours (≤72 h)**: **tenofovir 300 mg + emtricitabine 200 mg daily + raltegravir 400 mg bd × 28 days** — 1
- HBV: vaccine ± HBIG if non-immune; follow-up HIV testing at 6 weeks and 3 months — 1

Q2 What is the likely diagnosis in the patient, and how is HIV confirmed? **4 marks**

MARKING SCHEME

- ***Pneumocystis jirovecii* pneumonia** → AIDS-defining — 1
- Oral candidiasis → suspect HIV (CD4 likely <200) — 1
- HIV **ELISA (4th gen Ag/Ab)** → confirm with **Western blot** (or 2nd assay) — 1
- **CD4 count** + **viral load (PCR)**; induced sputum/BAL for PCP — 1

Q3 Define AIDS and give 4 opportunistic infections with their CD4 thresholds. **3 marks**

MARKING SCHEME

- AIDS = HIV + **CD4 <200** (<14%) or an AIDS-defining illness — 1
- <200: PCP, toxoplasmosis, cryptococcus, cryptosporidium — 1
- <50–75: MAC, **CMV retinitis**, CNS lymphoma; any count: TB, Kaposi — 1

Q4 Main lines of management for the patient. **5 marks**

MARKING SCHEME

- PCP: **high-dose co-trimoxazole** × 21 days — 1
- + **prednisolone** if PaO2 <70 mmHg — 1
- **ART for all**: 2 NRTIs (tenofovir + emtricitabine/lamivudine) + integrase inhibitor (**dolutegravir**), started within 2 weeks — 1
- Prophylaxis: co-trimoxazole (CD4 <200), azithromycin if CD4 <50–75 (MAC); screen TB, HBV/HCV, syphilis — 1
- Vaccines (no live vaccines if CD4 <200); partner notification; counselling & adherence — 1

DRUGS IN THIS CASE

Drug	Class	Key point
Tenofovir (TDF)	NRTI — nucleotide reverse-transcriptase inhibitor	Renal toxicity, bone loss
Emtricitabine / lamivudine	NRTIs	Also active against HBV
Raltegravir / dolutegravir	Integrase strand-transfer inhibitors (INSTI)	Backbone partner
Efavirenz	NNRTI	CNS side effects
Lopinavir/ritonavir	Protease inhibitor + booster	Many interactions
Co-trimoxazole	Sulphonamide + trimethoprim	PCP treatment & prophylaxis
Zidovudine	NRTI	Neonate PMTCT 6 weeks; anaemia

4 Haematology cases

8 cases · 41 questions

IDA vs thal · B12 · ITP · haemophilia/vWD · haemolysis · thalassaemia major · transfusion & DIC · thrombophilia

Case 32 Two young women with small red cells

A · 20 marks

Patient A: 22-year-old woman, easy fatigue, heavy menses for 2 years, pallor, brittle spoon-shaped nails. **Patient B**: 21-year-old woman, pre-employment check, slight pallor, no complaints.

A: Hb 8 · RBC 3.2 · MCV 67 · MCH 20 · RDW 20% · retics 0.5% · plt 400

B: Hb 10 · RBC 5.6 · MCV 54 · MCH 18 · RDW 12% · retics 1.5%

Q1 Describe the anaemia in each and give the most likely diagnosis. **4 marks**

Q2 Give the differential diagnosis of microcytic anaemia. **2 marks**

Q3 Which tests confirm each diagnosis? **4 marks**

Q4 Treat patient A with doses. **5 marks**

Q5 Should patient B take iron? Explain. **2 marks**

Q6 In a 60-year-old man with IDA, what must you look for? **3 marks**

Case 32 Two young women with small red cells

A · 20 marks

Q1 Describe the anaemia in each and give the most likely diagnosis. **4 marks**

MARKING SCHEME

- Both **microcytic hypochromic** anaemia — 1
- A: **iron-deficiency anaemia** (menorrhagia; ↑RDW, low RBC count, ↑platelets) — 1
- B: **β-thalassaemia trait** (mild anaemia, very low MCV, **high RBC count**, normal RDW) — 1
- Mentzer index MCV/RBC: A = 21 (>13 → IDA); B = 9.6 (<13 → thal trait) — 1

Q2 Give the differential diagnosis of microcytic anaemia. **2 marks**

MARKING SCHEME (½ EACH)

- Iron deficiency
- Thalassaemia trait / HbE
- Anaemia of chronic disease
- Sideroblastic anaemia (lead)

Q3 Which tests confirm each diagnosis? **4 marks**

MARKING SCHEME

- A: **serum ferritin ↓ (<12–15 µg/L)** — most sensitive & specific; ↓serum iron, ↑TIBC — 1
- A: ferritin unreliable with inflammation (acute-phase) → combine with iron/TIBC; blood film: pencil cells — 1
- B: **Hb electrophoresis — ↑HbA2 >3.5%** (± ↑HbF) — 1
- B: normal ferritin; film target cells; partner screening/genetic counselling — 1

Q4 Treat patient A with doses. **5 marks**

MARKING SCHEME

- **Treat the cause**: gynaecology review for menorrhagia (± tranexamic acid / hormonal therapy) — 1
- **Ferrous sulphate 200 mg (65 mg elemental iron) 2–3 times daily** (or once daily / alternate days) — 1
- Take on empty stomach / with vitamin C; S/E: GI upset, black stools — 1
- Expect retic peak ~7–10 days, Hb ↑ ~1–2 g/dL per 2–3 weeks; **continue 3 months after Hb normalises** — 1
- IV iron (ferric carboxymaltose) if intolerant/malabsorption/non-adherent; transfuse only if unstable — 1

Q5 Should patient B take iron? Explain. **2 marks**

MARKING SCHEME

- **No** — iron stores normal; unnecessary iron → overload — 1
- Folic acid in pregnancy; reassure; screen partner before pregnancy — 1

Q6 In a 60-year-old man with IDA, what must you look for? **3 marks**

MARKING SCHEME

- Occult **GI blood loss** — colon cancer, gastric cancer, peptic ulcer, NSAIDs — 1
- **Upper GI endoscopy + colonoscopy**; coeliac serology — 1
- Urine for haematuria; hookworm/bilharzia in endemic areas — 1

KEY COMPARISON · MICROCYTIC ANAEMIAS

	Iron deficiency	β-thal trait	Chronic disease
RBC count	↓	↑ (>5)	↓
RDW	↑	Normal	Normal
Ferritin	↓	N	N / ↑
TIBC	↑	N	↓
Key test	Ferritin	HbA2 electrophoresis	Treat disease

Case 33 Tingling feet and falls in the dark

A · 18 marks

A 55-year-old woman with vitiligo has 1 month of weakness and fatigue, tingling in both hands and feet, difficulty walking and repeated falls in the dark.

Pallor, lemon-yellow tint | Symmetrical ↓sensation, ↓vibration & position sense, +ve Romberg | Hb 8.6 · MCV 119 · retics 0.5% | WBC 3.5 · plt 135 | Film: macro-ovalocytes, hypersegmented neutrophils | LDH 3960

Q1 Describe the blood picture and give the diagnosis. **3 marks**

Q2 Investigations to confirm cause. **4 marks**

Q3 Give treatment with doses. **5 marks**

Q4 List causes of B12 and of folate deficiency. **4 marks**

Q5 How do you treat folate deficiency? **2 marks**

Case 33 Tingling feet and falls in the dark

A · 18 marks

Q1 Describe the blood picture and give the diagnosis. **3 marks**

MARKING SCHEME

- **Macrocytic (megaloblastic) anaemia** with pancytopenia; ↑LDH (ineffective erythropoiesis) 1
- **Vitamin B12 deficiency** — likely **pernicious anaemia** (autoimmune; vitiligo) 1
- Neurology = **subacute combined degeneration of the cord** + peripheral neuropathy 1

Q2 Investigations to confirm cause. **4 marks**

MARKING SCHEME

- **Serum B12** ↓ and **red-cell/serum folate** 1
- **Anti-intrinsic factor antibodies** (specific) / anti-parietal cell Ab 1
- ↑ methylmalonic acid & homocysteine; ↑indirect bilirubin 1
- Bone marrow (megaloblasts) if doubt; endoscopy (atrophic gastritis, gastric cancer risk), TSH 1

Q3 Give treatment with doses. **5 marks**

MARKING SCHEME

- **Hydroxocobalamin 1 mg IM** 1
- × **6 doses over 2 weeks** (3 times weekly); with neurology: alternate days until no further improvement 1
- Maintenance **1 mg IM every 3 months for life** (pernicious anaemia) 1
- **Never give folic acid alone** before B12 (precipitates/worsens SACD) 1
- Monitor K^+ (hypokalaemia in early response), retics day 5–7; iron may be needed 1

Q4 List causes of B12 and of folate deficiency. **4 marks**

MARKING SCHEME

- B12: **pernicious anaemia**, gastrectomy, terminal ileal resection/Crohn's 1
- B12: vegan diet, blind-loop/bacterial overgrowth, fish tapeworm, metformin/PPIs 1
- Folate: poor diet, alcohol, pregnancy, haemolysis 1
- Folate: coeliac disease, drugs (methotrexate, phenytoin, trimethoprim) 1

Q5 How do you treat folate deficiency? **2 marks**

MARKING SCHEME

- **Folic acid 5 mg orally daily × 4 months** 1
- After excluding/treating B12 deficiency; treat cause; long-term in chronic haemolysis 1

DRUGS IN THIS CASE

Drug	Class	Key point
Hydroxocobalamin	Vitamin B12 (haematinic)	IM; lifelong in pernicious anaemia
Folic acid	Vitamin B9	Not alone in B12 deficiency
Metformin	Biguanide antidiabetic	Causes B12 malabsorption
Methotrexate	Antimetabolite (DHFR inhibitor)	Folate deficiency → give folic acid

Case 34 Purple spots and nosebleeds

A · 18 marks

A 32-year-old woman has multiple petechiae, purpura and bruises without trauma, and recurrent epistaxis that needed cautery. She is otherwise well; no hepatosplenomegaly or lymphadenopathy.

Platelets 18 × 10^9/L | Bleeding time 10 min | PT 12 s · aPTT 32 s | Hb & WBC normal | Film: few large platelets, no schistocytes

Q1 Interpret the coagulation profile and give the diagnosis. **3 marks**

Q2 Which causes of thrombocytopenia must be excluded and how? **4 marks**

Q3 When and how do you treat? Give doses. **6 marks**

Q4 Compare ITP in children and adults. **2 marks**

Q5 A 19-year-old woman has long-standing epistaxis; platelets 180, BT 10 min. Name 2 possible diagnoses. **3 marks**

Case 34 Purple spots and nosebleeds

A · 18 marks

Q1 Interpret the coagulation profile and give the diagnosis. **3 marks**

MARKING SCHEME

- **Isolated thrombocytopenia** with ↑BT; normal PT & aPTT (coagulation factors normal) 1
- **Immune thrombocytopenic purpura (ITP)** — diagnosis of exclusion 1
- Anti-platelet antibodies → peripheral destruction (marrow megakaryocytes normal/↑) 1

Q2 Which causes of thrombocytopenia must be excluded and how? **4 marks**

MARKING SCHEME

- **Pseudothrombocytopenia** (EDTA clumping) — repeat in citrate, film 1
- Secondary immune: **SLE** (ANA), **HIV, HCV**, H. pylori, CLL/lymphoma, drugs (heparin, quinine) 1
- Consumption: **DIC, TTP** (schistocytes, ↑LDH, renal) 1
- Marrow failure/infiltration (aplastic, leukaemia, B12) — **marrow** if atypical/age >60; splenomegaly (sequestration) 1

Q3 When and how do you treat? Give doses. **6 marks**

MARKING SCHEME

- Treat if **platelets <20–30 × 10^9/L or bleeding** (she qualifies) 1
- First line: **prednisolone 1 mg/kg/day** × 2–4 weeks then taper (or dexamethasone 40 mg/day × 4 days) 1
- **IVIG 1 g/kg × 1–2 days** for rapid rise (serious bleeding, pre-surgery) 1
- Second line: **TPO-receptor agonists** (eltrombopag, romiplostim), rituximab 1
- **Splenectomy** for refractory chronic ITP (vaccinate ≥2 weeks before) 1
- **Platelet transfusion only for life-threatening bleeding**; avoid aspirin/NSAIDs/IM injections 1

Q4 Compare ITP in children and adults. **2 marks**

MARKING SCHEME

- Children: acute, after viral infection, M = F, usually self-limiting (observe) 1
- Adults: insidious, female, **chronic** (>12 months), often needs treatment 1

Q5 A 19-year-old woman has long-standing epistaxis; platelets 180, BT 10 min. Name 2 possible diagnoses. **3 marks**

MARKING SCHEME

- Normal count + ↑BT = **platelet function defect** 1
- **von Willebrand disease** 1
- **Glanzmann thrombasthenia** (no GPIIb/IIIa) — confirm by aggregation studies; also aspirin, uraemia 1

DRUGS IN THIS CASE

Drug	Class	Key point
Prednisolone / dexamethasone	Corticosteroids (immunosuppressants)	First line in ITP
IVIG	Pooled human immunoglobulin (Fc-receptor blockade)	Rapid platelet rise
Eltrombopag / romiplostim	Thrombopoietin-receptor agonists	Second line
Rituximab	Anti-CD20 monoclonal antibody	Second line; HBV reactivation
Tranexamic acid	Antifibrinolytic	Mucosal bleeding

Case 35 A swollen elbow after a small fall — and a girl with heavy periods

A · 20 marks

(a) A 3-year-old boy has a painful swollen left elbow after mild trauma; aspiration shows blood. His maternal uncle "bled a lot".
(b) A 13-year-old girl has severe menorrhagia since menarche and recurrent nosebleeds; her father bruises easily.

(a) Plt 190 · BT normal · PT normal · aPTT 68 s | (b) Plt 190 · BT 11 min · PT normal · aPTT 52 s

Q1 Most likely diagnosis in (a) and (b)? **4 marks**

Q2 Which specific tests confirm each? **4 marks**

Q3 Manage the boy's haemarthrosis. **5 marks**

Q4 Manage the girl's vWD. **4 marks**

Q5 Which test monitors heparin and which monitors warfarin? **3 marks**

Case 35 A swollen elbow after a small fall — and a girl with heavy periods

A · 20 marks

Q1 Most likely diagnosis in (a) and (b)? **4 marks**

MARKING SCHEME

- (a) **Haemophilia A** (FVIII deficiency) — or B (FIX) 1
- X-linked recessive (affected males, maternal uncle); haemarthrosis; only aPTT prolonged 1
- (b) **von Willebrand disease** (type 1) — autosomal dominant, commonest inherited bleeding disorder 1
- ↑BT (platelet adhesion) + ↑aPTT (↓FVIII carrier), mucosal bleeding 1

Q2 Which specific tests confirm each? **4 marks**

MARKING SCHEME

- (a) **FVIII and FIX assays**; mixing study corrects (factor deficiency, not inhibitor) 1
- (a) Severity: <1% severe, 1–5% moderate, 5–40% mild; screen for inhibitors 1
- (b) **vWF antigen** ↓, vWF activity (ristocetin cofactor) ↓, FVIII ↓ 1
- (b) **Defective ristocetin-induced platelet aggregation** 1

Q3 Manage the boy's haemarthrosis. **5 marks**

MARKING SCHEME

- **Factor VIII concentrate IV** immediately (recombinant/plasma-derived) 1
- Dose: **1 IU/kg raises FVIII ≈ 2%** → target 30–50% for joint bleed (≈25 IU/kg), 80–100% for major/surgery 1
- RICE: rest, ice, compression, elevation; analgesia with paracetamol (no NSAIDs/aspirin, no IM) 1
- Physiotherapy after; long-term **prophylactic factor** or emicizumab to prevent arthropathy 1
- Mild haemophilia A: **desmopressin 0.3 µg/kg**; cryoprecipitate/FFP only if concentrate unavailable 1

Q4 Manage the girl's vWD. **4 marks**

MARKING SCHEME

- **Desmopressin (DDAVP) 0.3 µg/kg IV/SC** (or intranasal) — releases vWF (type 1) 1
- **Tranexamic acid** for menorrhagia/epistaxis 1
- Combined OCP for menorrhagia; iron for IDA 1
- vWF-containing FVIII concentrate (or cryoprecipitate) for severe/type 3 or surgery 1

Q5 Which test monitors heparin and which monitors warfarin? **3 marks**

MARKING SCHEME

- UFH → **aPTT** (1.5–2.5 × control); LMWH → anti-Xa if needed 1
- Warfarin → **PT/INR** (target 2–3; 2.5–3.5 mechanical mitral) 1
- Samples: **3.2% sodium citrate, 1:9** ratio; deliver promptly 1

KEY CRITERIA · BLEEDING-TEST PATTERNS

Condition	Plt	BT	PT	aPTT	TT
ITP	↓	↑	N	N	N
Platelet function defect	N	↑	N	N	N
vWD	N	↑	N	↑	N
Haemophilia A/B	N	N	N	↑	N
Warfarin / vit K def / liver	N	N	↑↑	N/↑	N
Heparin	N	N	N/↑	↑↑	↑
DIC	↓	↑	↑	↑	↑ (+ ↓fibrinogen, ↑FDP)

DRUGS IN THIS CASE

Drug	Class	Key point
Factor VIII / IX concentrate	Coagulation factor replacement	Inhibitor development
Desmopressin (DDAVP)	Synthetic vasopressin (V2) analogue	Releases vWF & FVIII; hyponatraemia
Emicizumab	Bispecific monoclonal antibody (FVIII mimetic)	SC prophylaxis
Tranexamic acid	Antifibrinolytic	Mucosal bleeding
Cryoprecipitate	Blood component (FVIII, vWF, fibrinogen, FXIII)	Backup only

Case 36 Yellow, tired and passing red urine

A · 18 marks

A 44-year-old woman with SLE complains of fatigue and weakness. Her spleen is palpable.

Hb 8.5 · MCV 90 · retics 13% | Total bilirubin 4.2 (direct 0.3) | LDH ↑ · haptoglobin ↓ | Film: spherocytes, polychromasia | Urine: pinkish-red

Q1 What type of anaemia is this? Give the laboratory evidence. **4 marks**

Q2 What is the key next test and the likely diagnosis? **3 marks**

Q3 Treatment (doses where possible). **4 marks**

Q4 A 5-year-old boy with mild splenomegaly, MCHC 36.4, spherocytes, indirect bilirubin ↑, Coombs negative. Diagnosis and confirmation? **3 marks**

Q5 A boy develops jaundice and dark urine 2 days after eating fava beans. Diagnosis, film finding and when to test? **4 marks**

Case 36 Yellow, tired and passing red urine

A · 18 marks

Q1 What type of anaemia is this? Give the laboratory evidence. **4 marks**

MARKING SCHEME

- **Haemolytic anaemia** (normocytic, with reticulocytosis) 1
- ↑ **unconjugated (indirect) bilirubin**, ↑LDH, ↑urobilinogen 1
- ↓ **haptoglobin**; haemoglobinuria → intravascular component 1
- ↑ **retics**, polychromasia; spherocytes 1

Q2 What is the key next test and the likely diagnosis? **3 marks**

MARKING SCHEME

- **Direct antiglobulin (Coombs) test** 1
- Positive (IgG ± C3) → **warm autoimmune haemolytic anaemia** 1
- Secondary to **SLE** (others: CLL, lymphoma, viral, drugs — methyldopa) 1

Q3 Treatment (doses where possible). **4 marks**

MARKING SCHEME

- **Prednisolone 1 mg/kg/day** until Hb stable, then slow taper 1
- Second line: **rituximab**, splenectomy, azathioprine/mycophenolate 1
- **Folic acid 5 mg daily** (increased demand); treat SLE 1
- Transfuse only if severe/symptomatic, with best-matched blood (crossmatch difficult) 1

Q4 A 5-year-old boy with mild splenomegaly, MCHC 36.4, spherocytes, indirect bilirubin ↑, Coombs negative. Diagnosis and confirmation? **3 marks**

MARKING SCHEME

- **Hereditary spherocytosis** (autosomal dominant membrane defect) 1
- **Family study**, **osmotic fragility** ↑ / EMA binding test; Coombs −ve excludes AIHA 1
- Folate; **splenectomy** if severe (after age 6, vaccinate first); gallstones risk 1

Q5 A boy develops jaundice and dark urine 2 days after eating fava beans. Diagnosis, film finding and when to test? **4 marks**

MARKING SCHEME

- **G6PD deficiency** (X-linked) — acute intravascular haemolysis 1
- Film: **Heinz bodies**, bite/blister cells 1
- **Enzyme assay ~6 weeks after the attack** (falsely normal during haemolysis — young cells) 1
- Avoid oxidants: fava, primaquine, dapsone, sulphonamides, nitrofurantoin; transfuse if severe 1

KEY COMPARISON · INTRAVASCULAR VS EXTRAVASCULAR HAEMOLYSIS

	Intravascular	Extravascular
Site	Inside vessels	Spleen/liver macrophages
Examples	ABO mismatch, G6PD, PNH, malaria, MAHA	HS, warm AIHA, thalassaemia, sickle
Clues	Haemoglobinaemia, haemoglobinuria, haemosiderinuria, very ↓haptoglobin	Splenomegaly, spherocytes, jaundice, gallstones

DRUGS IN THIS CASE

Drug	Class	Key point
Prednisolone	Corticosteroid	First line warm AIHA
Rituximab	Anti-CD20 monoclonal antibody	Warm & cold AIHA
Azathioprine	Immunosuppressant (purine antimetabolite)	Steroid-sparing; check TPMT
Folic acid	Vitamin B9	Chronic haemolysis
Primaquine / dapsone	Antimalarial / sulphone	Oxidants — trigger G6PD haemolysis

Case 37 A pale child with a big spleen, ready for surgery

A · 18 marks

A 16-year-old boy has needed blood transfusions every 3–4 weeks since infancy. He has frontal bossing, maxillary overgrowth, massive splenomegaly and growth delay. His transfusion needs have increased and splenectomy is planned.

Hb 6.5 pre-transfusion | MCV 60 | Film: target cells, normoblasts, anisopoikilocytosis | Ferritin 3500 µg/L | HbF 85%, HbA absent

Q1 Diagnosis and pathogenesis. **3 marks**

Q2 Outline the transfusion and chelation plan. **5 marks**

Q3 Which vaccines before splenectomy, and when? **4 marks**

Q4 List complications of iron overload. **3 marks**

Q5 What is the only curative option and how is the disease prevented? **3 marks**

Case 37 A pale child with a big spleen, ready for surgery

A · 18 marks

Q1 Diagnosis and pathogenesis. **3 marks**

MARKING SCHEME

- **β-thalassaemia major** (β0/β0) — 1
- Absent β-chain → excess α-chains precipitate → **ineffective erythropoiesis + haemolysis** — 1
- Marrow expansion (skull/facial bones), extramedullary haemopoiesis (HSM) — 1

Q2 Outline the transfusion and chelation plan. **5 marks**

MARKING SCHEME

- Regular transfusion every 2–4 weeks to keep **pre-transfusion Hb 9–10.5 g/dL** — 1
- Leucodepleted, phenotype-matched red cells; check for alloantibodies — 1
- **Iron chelation** when ferritin >1000 µg/L (or after ~10–20 transfusions) — 1
- **Desferrioxamine SC infusion** 8–12 h, 5–7 nights/week, or oral **deferasirox** / deferiprone — 1
- Monitor ferritin, cardiac MRI T2*, LFT, endocrine (DM, hypogonadism, hypothyroid); folic acid — 1

Q3 Which vaccines before splenectomy, and when? **4 marks**

MARKING SCHEME

- **Pneumococcal** vaccine — 1
- ***Haemophilus influenzae*** **type b** + **meningococcal (ACWY + B)** — 1
- Give **≥2 weeks before** elective splenectomy (ideally 4–6 weeks); emergency → ≥14 days after — 1
- + annual influenza vaccine and lifelong **penicillin V prophylaxis**; emergency antibiotics card — 1

Q4 List complications of iron overload. **3 marks**

MARKING SCHEME (½ EACH)

- Cardiomyopathy / HF, arrhythmias (main cause of death)
- Liver fibrosis / cirrhosis
- Diabetes mellitus
- Hypogonadism, growth failure
- Hypothyroidism, hypoparathyroidism
- Skin pigmentation

Q5 What is the only curative option and how is the disease prevented? **3 marks**

MARKING SCHEME

- **Allogeneic haematopoietic stem-cell (bone-marrow) transplant** from matched sibling — 1
- Gene therapy emerging — 1
- Prevention: carrier screening (premarital), genetic counselling, prenatal diagnosis — 1

DRUGS IN THIS CASE

Drug	Class	Key point
Desferrioxamine	Iron chelator (parenteral)	Auditory/visual toxicity, Yersinia
Deferasirox	Oral iron chelator	Renal & hepatic toxicity
Deferiprone	Oral iron chelator	Agranulocytosis — weekly CBC
Penicillin V	Narrow-spectrum penicillin	Post-splenectomy prophylaxis
Pneumococcal / Hib / meningococcal vaccines	Inactivated conjugate/polysaccharide vaccines	Encapsulated organisms
Folic acid	Vitamin B9	↑ erythropoietic demand

Case 38 Twenty minutes into a transfusion

A · 20 marks

A 50-year-old woman receiving her first unit of packed red cells after surgery develops fever, rigors, loin pain, chest tightness and hypotension 20 minutes into the transfusion. Later her urine turns dark red and she bleeds from cannula sites.

BP 80/50 | Temp 39 | Plt 40,000 | PT & aPTT prolonged | Fibrinogen ↓ | D-dimer ↑↑ | Film: schistocytes

Q1 What happened first? What is the commonest reason? **3 marks**

Q2 Immediate management steps. **5 marks**

Q3 What complication developed next? Interpret the labs and treat. **5 marks**

Q4 Give 4 other causes of DIC. **2 marks**

Q5 Match these reactions to their timing/feature: febrile non-haemolytic, anaphylaxis, TRALI, delayed haemolytic, TA-GVHD. **5 marks**

Case 38 Twenty minutes into a transfusion

A · 20 marks

Q1 What happened first? What is the commonest reason? **3 marks**

MARKING SCHEME

- **Acute haemolytic transfusion reaction** (within first 30 min) — 1
- Due to **ABO incompatibility** — naturally occurring anti-A/anti-B → intravascular haemolysis — 1
- Usually **clerical/identification error** (wrong patient/sample) — 1

Q2 Immediate management steps. **5 marks**

MARKING SCHEME

- **STOP the transfusion immediately** — 1
- Keep IV line open with **0.9% saline**; resuscitate, maintain urine output >100 mL/h — 1
- **Re-check patient identity vs unit** at bedside — 1
- Notify physician & **blood bank**; return the unit + giving set — 1
- Send fresh blood (repeat group, DAT, CBC, bilirubin, LDH, coagulation, renal) & **urine** (haemoglobinuria) — 1

Q3 What complication developed next? Interpret the labs and treat. **5 marks**

MARKING SCHEME

- **Disseminated intravascular coagulation** — 1
- ↓platelets, ↑PT/aPTT/TT, ↓fibrinogen, ↑FDP/D-dimer, schistocytes — 1
- **Treat the cause** + supportive care — 1
- Bleeding: **platelets** if <50, **FFP** (factors), **cryoprecipitate** if fibrinogen <1.5 g/L — 1
- Watch AKI (fluids, dialysis); heparin only if thrombosis predominates — 1

Q4 Give 4 other causes of DIC. **2 marks**

MARKING SCHEME (½ EACH)

- Sepsis (Gram −ve, meningococcal), falciparum malaria
- Malignancy — APL (M3), prostate, adenocarcinoma
- Obstetric — abruption, amniotic embolism, eclampsia, septic abortion
- Trauma, burns, snake bite

Q5 Match these reactions to their timing/feature: febrile non-haemolytic, anaphylaxis, TRALI, delayed haemolytic, TA-GVHD. **5 marks**

MARKING SCHEME (1 EACH)

- **FNHTR**: 30–90 min, fever/chills, anti-HLA → paracetamol; prevented by leucodepletion
- **Anaphylaxis**: minutes; **IgA-deficient recipient** → adrenaline; future washed cells
- **TRALI**: within 6 h, ARDS-like, donor anti-leucocyte Ab
- **Delayed haemolytic**: 5–10 days, extravascular (Kidd, Duffy, Kell, Rh)
- **TA-GVHD**: 4–30 days, rash, liver, GI, marrow failure — fatal; prevent by **irradiation**

DRUGS IN THIS CASE

Drug / product	Class	Key point
0.9% saline	Isotonic crystalloid	Only fluid compatible with blood
Fresh frozen plasma	Blood component — all coagulation factors	Stored −18/−30 °C
Cryoprecipitate	Blood component — fibrinogen, FVIII, vWF, FXIII	Hypofibrinogenaemia
Platelet concentrate	Blood component	20–24 °C, agitated, 5 days
Paracetamol	Antipyretic	FNHTR
Adrenaline	Sympathomimetic	Anaphylactic reaction
Furosemide	Loop diuretic	TACO (circulatory overload)

Case 39 A third clot at 29

A · 16 marks

A 29-year-old woman presents with her second left-leg DVT; her brother had a PE at 35. She has had 3 first-trimester miscarriages. No obvious provoking factor now.

aPTT 48 s, not corrected by mixing | Plt 120 | PT normal | Doppler: femoral DVT

Q1 Who should be screened for thrombophilia? Does she qualify? **3 marks**

Q2 Which tests make up the panel? **4 marks**

Q3 The prolonged aPTT not corrected by mixing suggests what? Diagnosis? **3 marks**

Q4 Management of the DVT and long-term plan. **4 marks**

Q5 Explain what each natural anticoagulant does. **2 marks**

Case 39 A third clot at 29

A · 16 marks

Q1 Who should be screened for thrombophilia? Does she qualify? **3 marks**

MARKING SCHEME

- Unprovoked VTE <50 years; family history of VTE/known thrombophilia — 1
- Recurrent VTE; unusual sites (mesenteric, cerebral, portal); **recurrent miscarriage → lupus anticoagulant** — 1
- She qualifies on several criteria — 1

Q2 Which tests make up the panel? **4 marks**

MARKING SCHEME

- CBC + film (MPN, PNH) — 1
- **Lupus anticoagulant**, anticardiolipin & anti-β2-glycoprotein I antibodies — 1
- **Protein C, free protein S, antithrombin III** (off anticoagulants, not during acute clot) — 1
- **Factor V Leiden**, **prothrombin G20210A**, homocysteine/MTHFR — 1

Q3 The prolonged aPTT not corrected by mixing suggests what? Diagnosis? **3 marks**

MARKING SCHEME

- A **circulating inhibitor** — lupus anticoagulant (paradoxically pro-thrombotic) — 1
- **Antiphospholipid syndrome**: thrombosis/pregnancy morbidity + persistent antibodies (≥12 weeks apart) — 1
- Look for SLE (ANA, dsDNA) — 1

Q4 Management of the DVT and long-term plan. **4 marks**

MARKING SCHEME

- LMWH (enoxaparin 1 mg/kg bd) overlapped with **warfarin, INR 2–3** — 1
- APS (esp. triple-positive): **warfarin preferred over DOACs** — 1
- Recurrent/unprovoked + APS → **long-term (indefinite)** anticoagulation — 1
- Avoid oestrogen OCP; pregnancy → stop warfarin (teratogenic), **LMWH + low-dose aspirin** — 1

Q5 Explain what each natural anticoagulant does. **2 marks**

MARKING SCHEME

- **Antithrombin III** inactivates thrombin (& Xa) — potentiated by heparin (AT deficiency → heparin resistance) — 1
- **Protein C** (activated by thrombin) + cofactor **protein S** inactivate FVa and FVIIIa; FV Leiden resists activated protein C — 1

💊 DRUGS IN THIS CASE

Drug	Class	Key point
Enoxaparin	LMWH anticoagulant	Safe in pregnancy
Warfarin	Vitamin K antagonist	Preferred in APS; teratogenic
Rivaroxaban / apixaban	DOACs (direct Xa inhibitors)	Avoid in triple-positive APS
Low-dose aspirin	Antiplatelet (COX-1 inhibitor)	APS in pregnancy
Vitamin K (phytomenadione)	Warfarin reversal agent	With PCC/FFP for bleeding

5 Immunology cases

3 cases · 16 questions | A: anaphylaxis & type I hypersensitivity · angioedema · drug allergy · allergic airway disease | B: hypersensitivity types 2–5 · chronic urticaria

Case 40 Five minutes after the injection

A · 22 marks

A 25-year-old man receives an IM injection of benzathine penicillin in a clinic. Within 5 minutes he develops generalised itching, urticaria, swelling of the lips and tongue, hoarseness, wheeze and dizziness, then collapses.

BP 70/40 | HR 135 | RR 30 | SpO2 88% | Stridor and bilateral wheeze

Q1 What is the diagnosis and the immunological mechanism? **3 marks**

Q2 Give the immediate management in order, with doses. **8 marks**

Q3 Which investigation supports the diagnosis afterwards? **2 marks**

Q4 Before discharge, what must be arranged? **4 marks**

Q5 His rheumatic-fever prophylaxis must continue. What do you give instead? **2 marks**

Q6 Classify hypersensitivity reactions with an example of each. **3 marks**

Case 40 Five minutes after the injection

A · 22 marks

Q1 What is the diagnosis and the immunological mechanism? **3 marks**

MARKING SCHEME

- **Anaphylaxis / anaphylactic shock** to penicillin — 1
- **Type I (IgE-mediated) hypersensitivity**: allergen cross-links IgE on mast cells/basophils — 1
- Release of histamine, tryptase, leukotrienes → vasodilatation, ↑permeability, bronchospasm — 1

Q2 Give the immediate management in order, with doses. **8 marks**

MARKING SCHEME

- Call for help; remove trigger; ABC — 1
- **Adrenaline 0.5 mg IM (0.5 mL of 1:1000)** into anterolateral thigh — lecture version: 1:1000, 1 mL SC — 1
- **Repeat every 5 minutes** if no improvement; IV adrenaline infusion only by experienced staff — 1
- Lie flat with legs raised (sit up if breathing difficulty predominates) — 1
- **High-flow O2**; secure airway early (stridor → anaesthetist, intubation/cricothyroidotomy) — 1
- **IV crystalloid 500–1000 mL bolus** (repeat as needed) — 1
- Then **chlorphenamine 10 mg IV/IM** + **hydrocortisone 200 mg IV** (adjuncts, not first) — 1
- Nebulised **salbutamol 5 mg** for persistent bronchospasm; glucagon if on β-blockers — 1

Q3 Which investigation supports the diagnosis afterwards? **2 marks**

MARKING SCHEME

- **Serum mast-cell tryptase** — ideally 1–2 h after onset (and baseline >24 h later) — 1
- Later: specific IgE / skin-prick testing by allergist (not during reaction) — 1

Q4 Before discharge, what must be arranged? **4 marks**

MARKING SCHEME

- Observe ≥6–12 h (risk of **biphasic reaction**) — 1
- Prescribe **2 adrenaline auto-injectors** + teach use — 1
- Document **penicillin allergy** in notes; medical alert bracelet — 1
- Refer to allergy/immunology clinic; written action plan — 1

Q5 His rheumatic-fever prophylaxis must continue. What do you give instead? **2 marks**

MARKING SCHEME

- **Erythromycin 250 mg bd** orally (macrolide) — 1
- Avoid all penicillins; cephalosporins only with specialist advice (cross-reactivity ~2%) — 1

Q6 Classify hypersensitivity reactions with an example of each. **3 marks**

MARKING SCHEME (½–1 EACH)

- **Type I** IgE immediate — anaphylaxis, urticaria, asthma, hay fever
- **Type II** antibody vs cell surface — AIHA, transfusion reaction, Goodpasture, ITP; (type V stimulating — Graves, inhibiting — myasthenia)
- **Type III** immune complex — serum sickness, SLE nephritis, post-strep GN, urticarial vasculitis
- **Type IV** delayed T-cell — contact dermatitis, TB skin test, SJS/TEN, papular urticaria

KEY CRITERIA · ANAPHYLAXIS (CLINICAL DIAGNOSIS)

Sudden onset after exposure	Airway	Breathing	Circulation	Skin / mucosa
Minutes (IV/IM drugs, stings) to ~1 h (food)	Tongue/throat swelling, hoarseness, stridor	Wheeze, dyspnoea, SpO2 <92%	Hypotension, tachycardia, collapse	Urticaria, flushing, angioedema (absent in ~20%)

Case 41 Swollen lips without the itch

A · 18 marks

(a) A 58-year-old hypertensive man started on **enalapril** 3 months ago presents with marked swelling of the lips and tongue, no urticaria, no itch. **(b)** A 20-year-old woman has had recurrent episodes since adolescence of non-itchy swelling of the hands, face and abdominal pain attacks, often after dental work; her mother has the same. Antihistamines never helped.

Q1 Define angioedema and how it differs from urticaria. **3 marks**

Q2 (a) What is the cause and mechanism? How do you manage it? **5 marks**

Q3 (b) Most likely diagnosis and confirmatory tests? **4 marks**

Q4 (b) Treatment of an acute attack and prophylaxis. **4 marks**

Q5 Name 2 other drugs that cause angioedema. **2 marks**

Case 41 Swollen lips without the itch

A · 18 marks

Q1 Define angioedema and how it differs from urticaria. **3 marks**

MARKING SCHEME

- Swelling of **deep dermis/subcutaneous & submucosal** tissue (lips, eyelids, tongue, genitalia, larynx, gut) 1
- Urticaria: superficial dermal **itchy wheals lasting <24 h** 1
- Angioedema: less itchy/painful, lasts up to 72 h; **laryngeal oedema is life-threatening** 1

Q2 (a) What is the cause and mechanism? How do you manage it? **5 marks**

MARKING SCHEME

- **ACE-inhibitor–induced angioedema** (can occur months–years after starting) 1
- **Bradykinin-mediated** (ACE normally degrades bradykinin) — not histamine 1
- Airway assessment first; anaesthetic support if stridor/voice change 1
- **Stop the ACE inhibitor permanently** (avoid all ACEi; ARB with caution) 1
- Antihistamines/steroids/adrenaline often ineffective; icatibant or FFP in severe cases 1

Q3 (b) Most likely diagnosis and confirmatory tests? **4 marks**

MARKING SCHEME

- **Hereditary angioedema** (C1-esterase inhibitor deficiency, autosomal dominant) 1
- Bradykinin-mediated: no urticaria, no itch, **no response to antihistamines**; trauma/dental work triggers 1
- **Low C4** (best screening test, even between attacks) 1
- **Low C1-INH level or function** (type I / type II) 1

Q4 (b) Treatment of an acute attack and prophylaxis. **4 marks**

MARKING SCHEME

- Acute: **C1-INH concentrate IV** 1
- or **icatibant** (bradykinin B2 antagonist) SC; FFP if unavailable 1
- Short-term prophylaxis before dental/surgical procedures: C1-INH (or danazol/tranexamic acid) 1
- Long-term: lanadelumab / berotralstat or danazol; avoid ACEi and oestrogens 1

Q5 Name 2 other drugs that cause angioedema. **2 marks**

MARKING SCHEME (1 EACH)

- NSAIDs/aspirin, penicillins, ARBs (rarely), gliptins, oestrogens (HAE)

KEY COMPARISON · HISTAMINE VS BRADYKININ ANGIOEDEMA

	Histamine (allergic)	Bradykinin (ACEi, HAE)
Urticaria / itch	Usually present	Absent
Onset	Minutes	Hours, lasts 2–5 days
Response to adrenaline/antihistamine/steroid	Good	Poor
Specific therapy	Adrenaline, antihistamine	C1-INH concentrate, icatibant; stop ACEi
Lab	Tryptase ↑ (anaphylaxis)	HAE: C4 ↓, C1-INH ↓

Case 42 Itchy wheals for two months — and a runny nose every spring

B / A · 18 marks

A 30-year-old woman has had daily itchy wheals all over the body for 2 months; each wheal fades within hours without a mark. She takes ibuprofen often for headaches. She also has sneezing, itchy watery eyes and nasal blockage every spring, and mild asthma.

Q1 Diagnosis and classification of her rash? **3 marks**

Q2 List causes/triggers of urticaria (immune and non-immune). **3 marks**

Q3 Investigations? **3 marks**

Q4 Treatment ladder for chronic urticaria with doses. **5 marks**

Q5 Diagnose her seasonal symptoms and give treatment. **4 marks**

Case 42 Itchy wheals for two months — and a runny nose every spring

B / A · 18 marks

Q1 Diagnosis and classification of her rash? **3 marks**

MARKING SCHEME

- **Chronic spontaneous urticaria** — wheals recurring for **>6 weeks** — 1
- Acute urticaria = <6 weeks — 1
- Each wheal <24 h and leaves no mark (if >24 h with bruising → **urticarial vasculitis** → biopsy) — 1

Q2 List causes/triggers of urticaria (immune and non-immune). **3 marks**

MARKING SCHEME

- Immune (type I IgE): foods, drugs (penicillin), insect stings, inhalants, infections, infestations — 1
- Non-immune: **NSAIDs/aspirin** (her ibuprofen!), opioids, food additives, contrast — 1
- Physical (cold, heat, pressure, sun, cholinergic, dermographism); autoimmune (thyroid), H. pylori — 1

Q3 Investigations? **3 marks**

MARKING SCHEME

- Mainly **clinical**; extensive tests rarely helpful — 1
- CBC (eosinophils), CRP/ESR, **TSH & anti-TPO** — 1
- Total IgE; biopsy only if lesions >24 h / vasculitis suspected; stool for parasites if indicated — 1

Q4 Treatment ladder for chronic urticaria with doses. **5 marks**

MARKING SCHEME

- **Avoid triggers** — stop NSAIDs (use paracetamol) — 1
- Step 1: **second-generation H1 antihistamine** daily, e.g. cetirizine/loratadine/fexofenadine **10 mg** (180 mg fexofenadine) — 1
- Step 2: **up-dose up to 4×** standard dose after 2–4 weeks — 1
- Step 3: **omalizumab** (anti-IgE) 300 mg SC every 4 weeks — 1
- Step 4: **ciclosporin**; short oral steroid course only for severe exacerbations; montelukast add-on — 1

Q5 Diagnose her seasonal symptoms and give treatment. **4 marks**

MARKING SCHEME

- **Seasonal allergic rhinoconjunctivitis** (pollen, type I) — allergic airway disease with asthma ("one airway") — 1
- Allergen avoidance; **intranasal corticosteroid** (mometasone/fluticasone 2 sprays each nostril daily) — most effective — 1
- Non-sedating oral antihistamine ± antihistamine eye drops; montelukast (helps asthma too) — 1
- Skin-prick/specific IgE to confirm allergen; **allergen immunotherapy** (level C) if refractory — 1

DRUGS IN THIS CASE

Drug	Class	Key point
Cetirizine / loratadine / fexofenadine	Second-generation (non-sedating) H1 antihistamines	Up-dose ×4 in chronic urticaria
Chlorphenamine / hydroxyzine	First-generation (sedating) H1 antihistamines	Night-time itch
Omalizumab	Anti-IgE monoclonal antibody	Refractory chronic urticaria / allergic asthma
Ciclosporin	Calcineurin inhibitor (immunosuppressant)	Nephrotoxic, ↑BP
Montelukast	Leukotriene-receptor antagonist	Rhinitis + asthma; mood changes
Mometasone / fluticasone nasal	Intranasal corticosteroids	First line allergic rhinitis
Ibuprofen	NSAID (COX inhibitor)	Non-immune urticaria trigger